

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short
portfolio, net of 20 bp transaction cost.
225 factors over 104 months, ordered by CAGR.

HOW TO READ

Each curve starts at 0 % in December 2017 and
compounds monthly long–short returns. The line is
coloured by the factor's style; the figure at the
right edge is the period-end cumulative return.

FACTOR STYLES

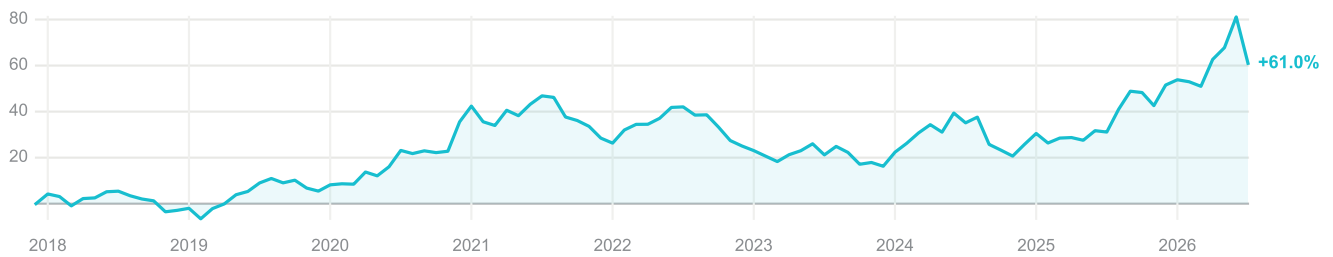
● Valuation	35	● Price Momentum	50
● Earnings Quality	46	● Size	4
● Analyst Expectations	18	● Historical Growth	38
● Capital Efficiency	27	● Volatility	7

001 In current model portfolio (28 factors)

001 ● 3M Revision in FY2 EPS Estimate

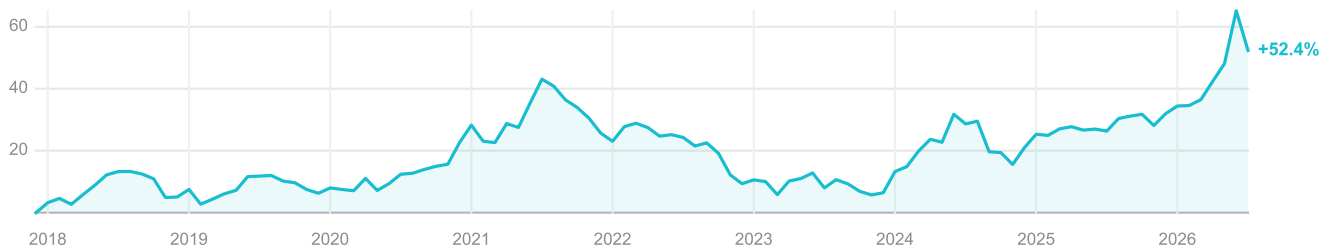
Analyst Expectations

CAGR +7.4%

**002 ● Street Revision Magnitude**

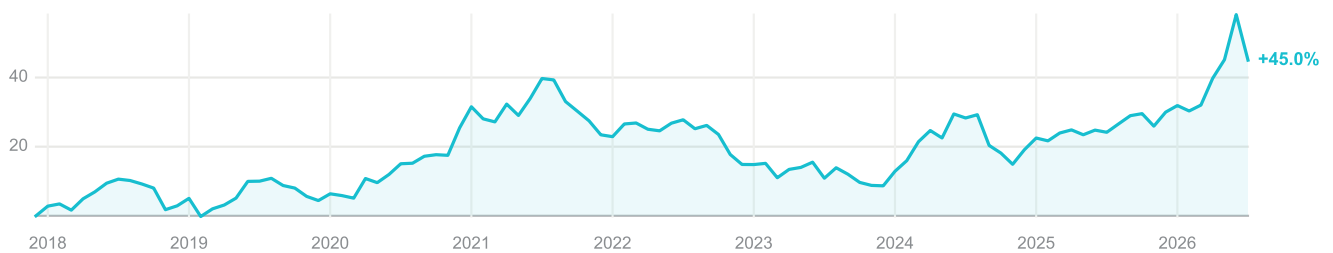
Analyst Expectations

CAGR +7.1%

**003 ● 3M Revision in FY1 EPS Estimate**

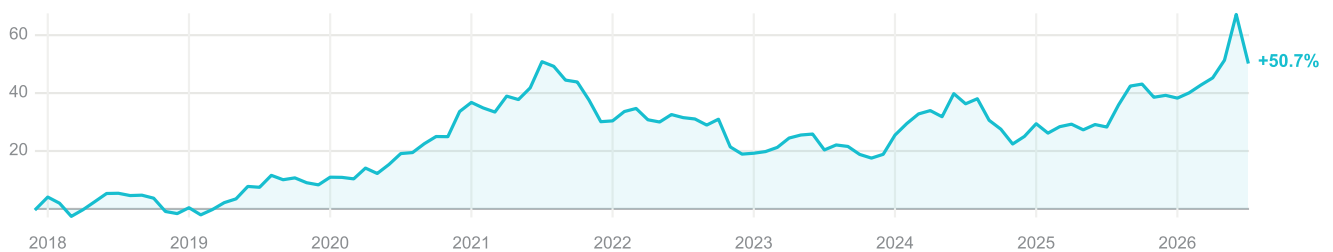
Analyst Expectations

CAGR +6.9%

**004 ● Number of EPS FY1 Revisions**

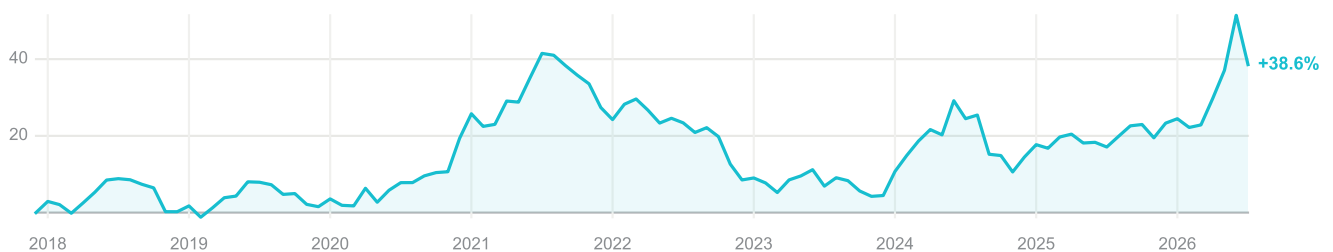
Analyst Expectations

CAGR +6.6%

**005 ● Adjusted Revision Magnitude**

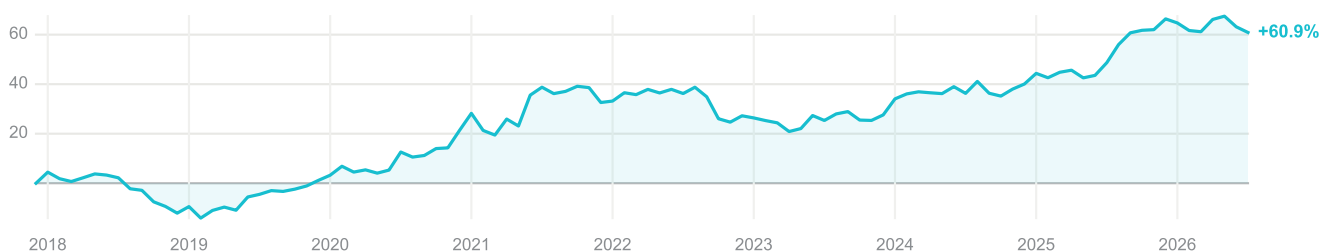
Analyst Expectations

CAGR +6.2%

**006 ● Standardized Unexpected Earnings**

Analyst Expectations

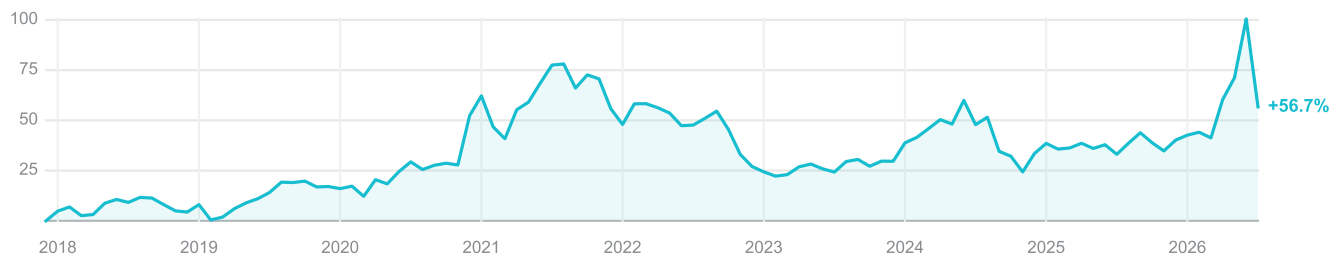
CAGR +5.9%



007 ● 6M Chg in Target Price

Analyst Expectations

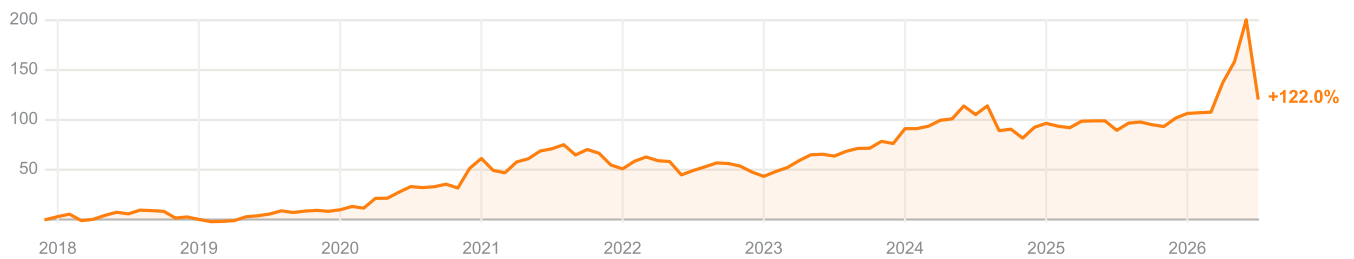
CAGR +5.8%



008 ● Slope of 52W Price Trend

Price Momentum

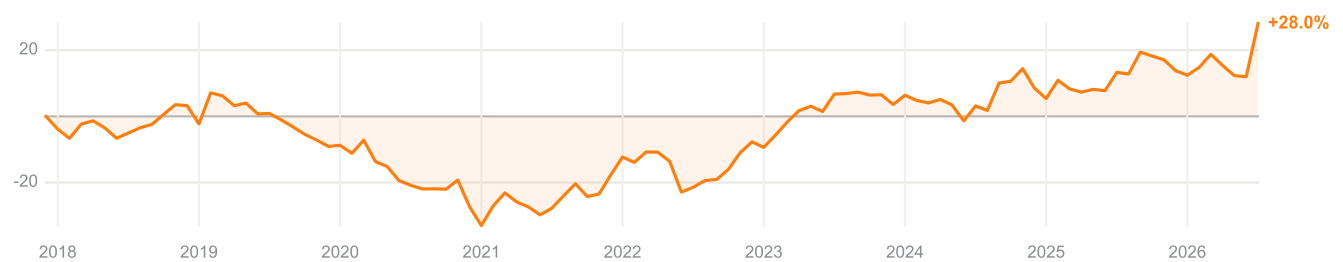
CAGR +5.7%



009 ● Sharpe Ratio

Price Momentum

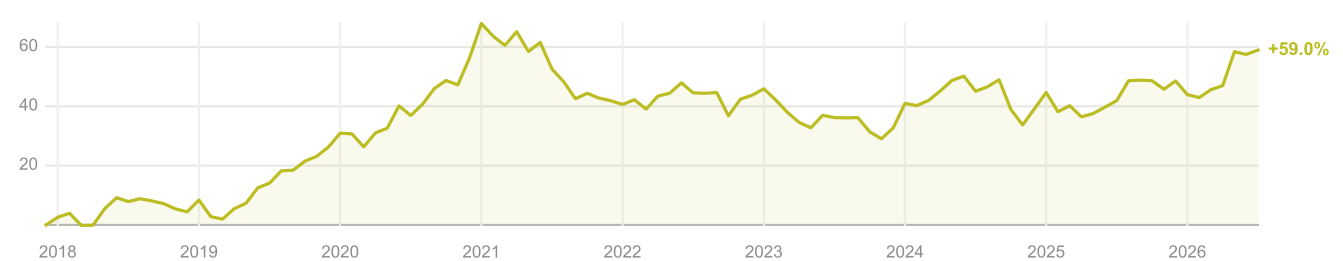
CAGR +5.3%



010 ● Cash Flow Return on Invested Capital

Capital Efficiency

CAGR +5.2%



011 ● Operating Cash Flow to Enterprise Value

Valuation

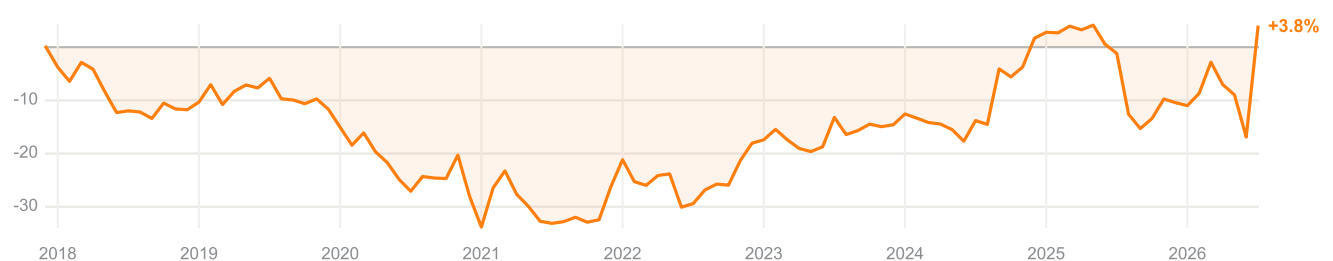
CAGR +5.0%



012 ● Closing Price to 260 Day Low

Price Momentum

CAGR +5.0%



013 ● Dividend per Share Growth

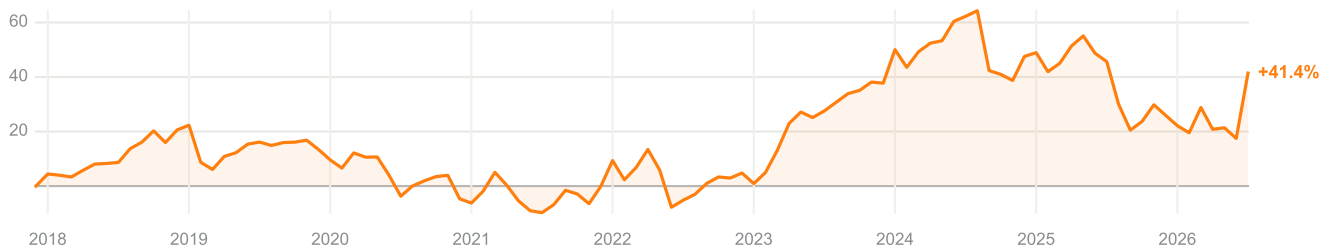
Historical Growth

CAGR +5.0%

**014 ● 24M Residual Return Variance**

Price Momentum

CAGR +5.0%

**015 ● 3Y Chg in Earnings per Share**

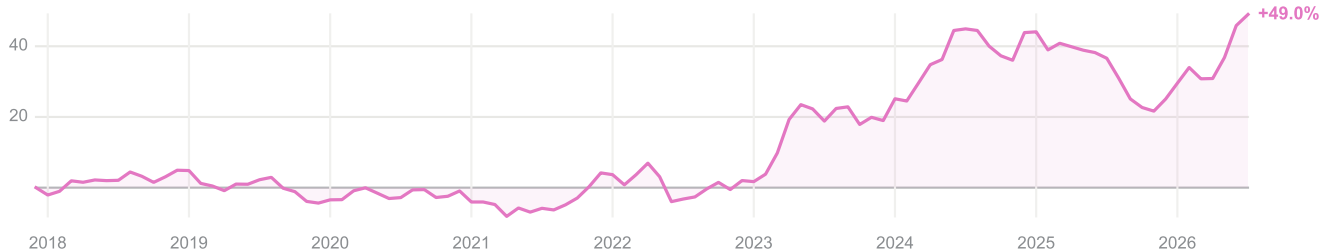
Historical Growth

CAGR +4.8%

**016 ● ROE 20 Qtr Standard Deviation**

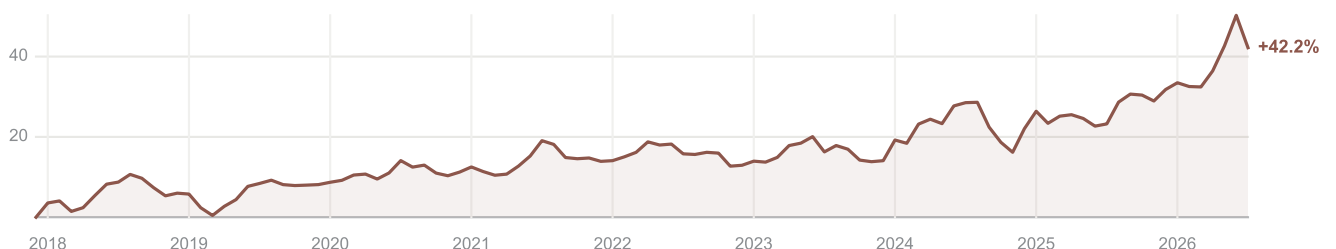
Earnings Quality

CAGR +4.8%

**017 ● 1Y Chg in Sales Turnover**

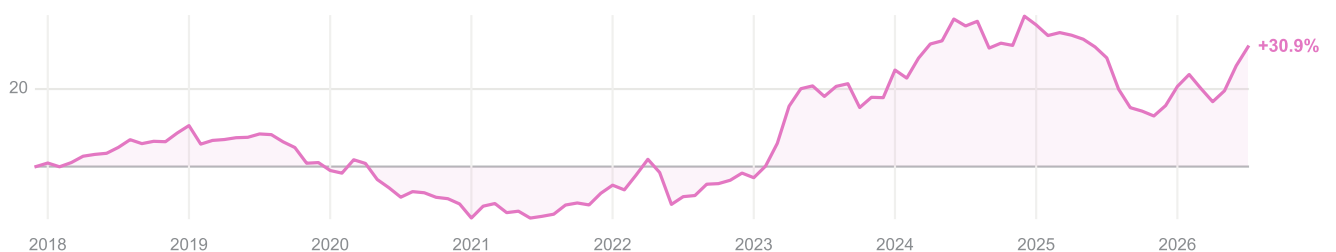
Historical Growth

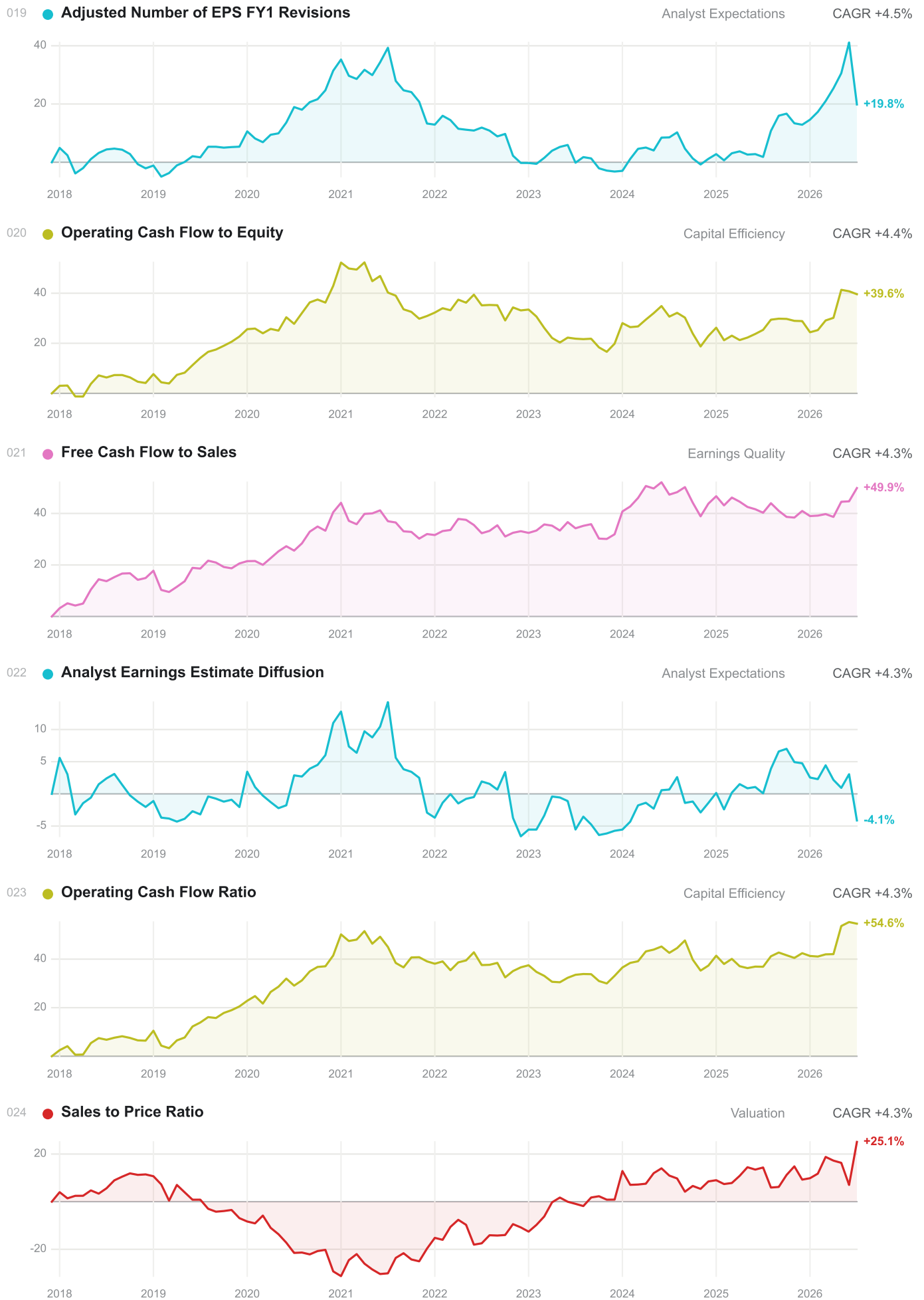
CAGR +4.7%

**018 ● ROA 20 Qtr Standard Deviation**

Earnings Quality

CAGR +4.5%

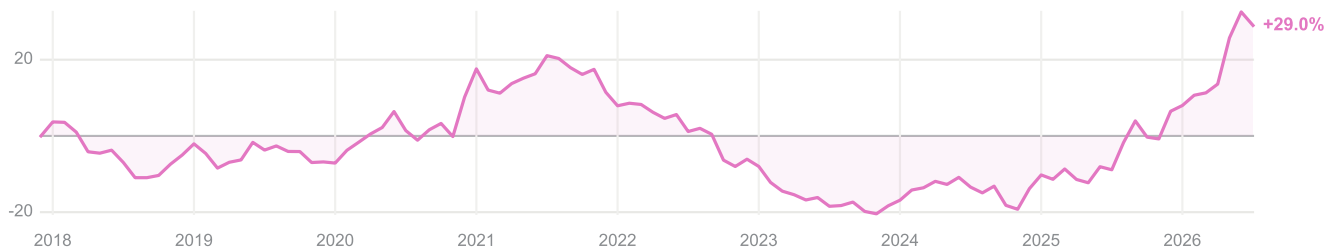




025 ● Net Profit Margin

Earnings Quality

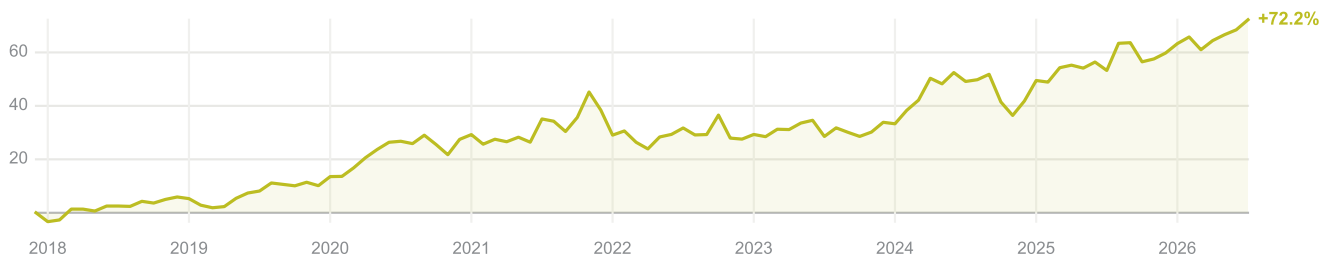
CAGR +4.2%



026 ● Long Term Debt to Equity Ratio

Capital Efficiency

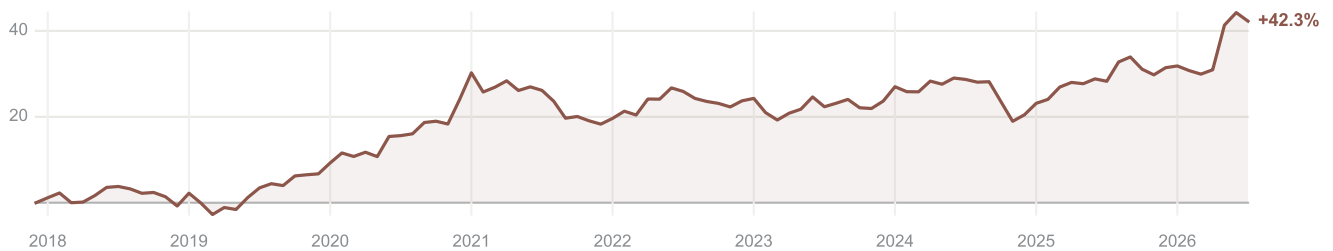
CAGR +4.2%



027 ● 1Y Chg in Operating Cash Flow per Share

Historical Growth

CAGR +4.2%



028 ● Free Cash Flow to Common Equity

Valuation

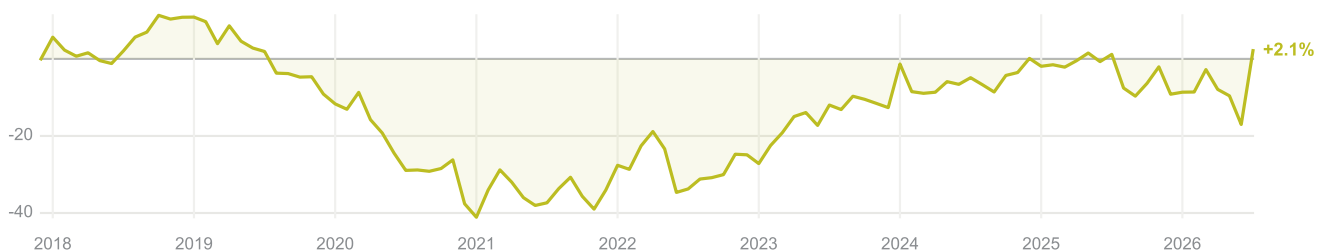
CAGR +3.9%



029 ● Tobin's Q Ratio

Capital Efficiency

CAGR +3.9%



030 ● Operating Cash Flow to Price

Valuation

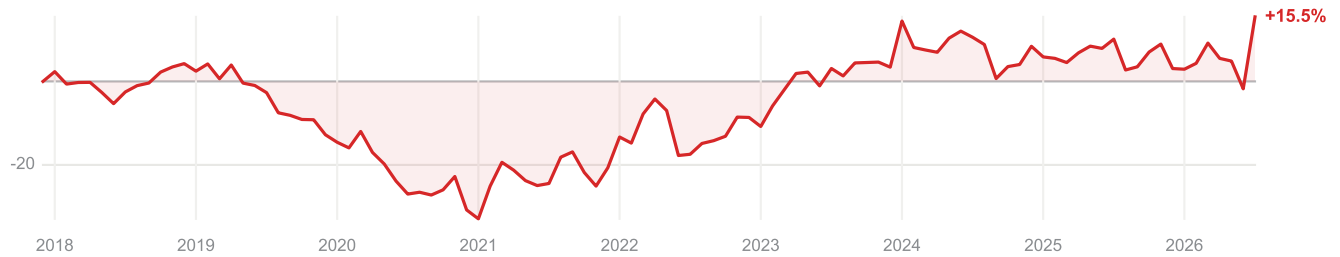
CAGR +3.8%



031 ● **Current Liability to Price**

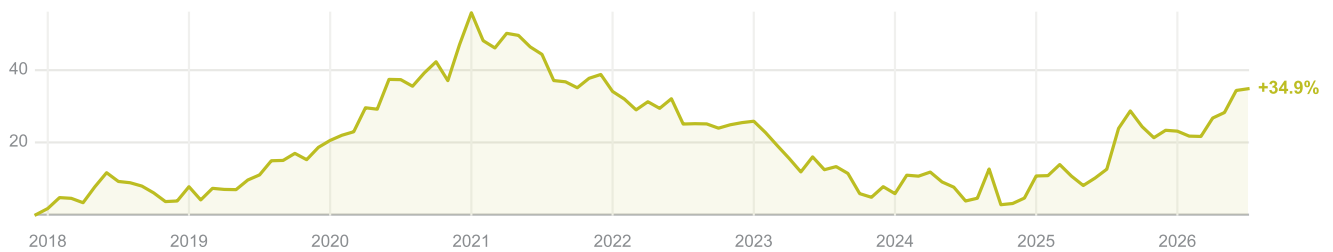
Valuation

CAGR +3.8%

032 ● **Gross Profit to Assets**

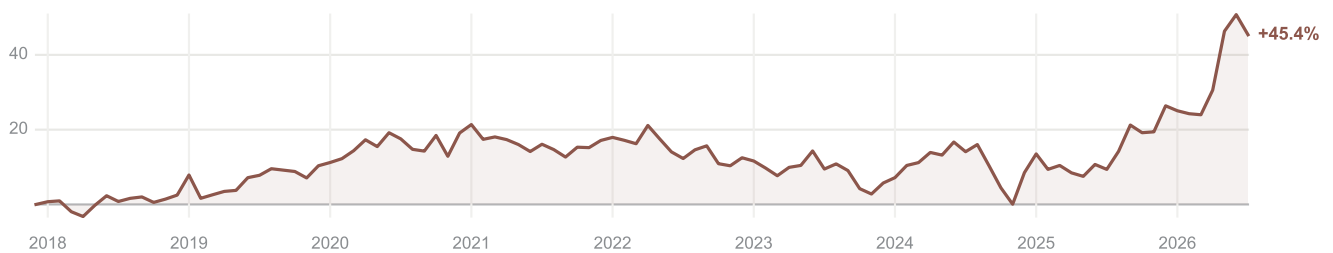
Capital Efficiency

CAGR +3.7%

033 ● **3Y Chg in Operating Margin**

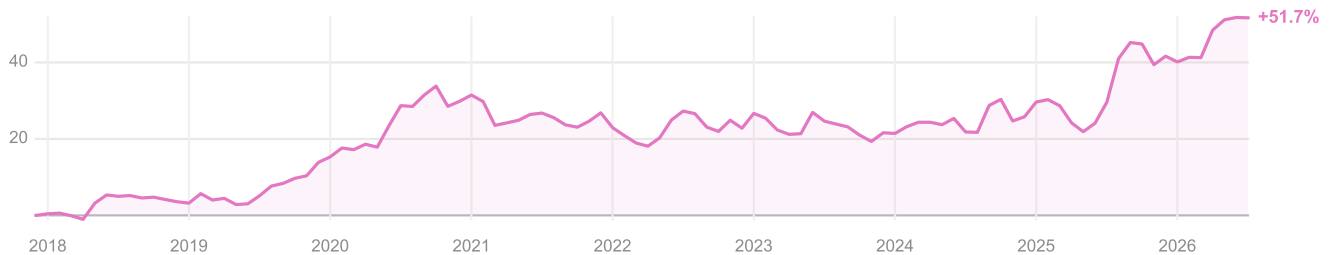
Historical Growth

CAGR +3.7%

034 ● **Asset Turnover**

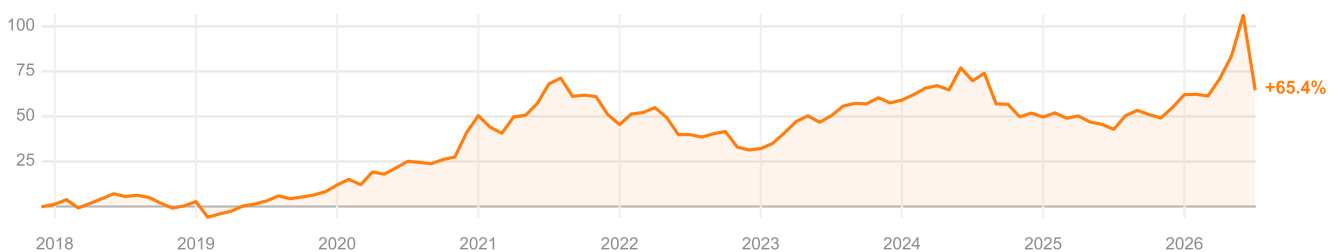
Earnings Quality

CAGR +3.7%

035 ● **180D Price TStat**

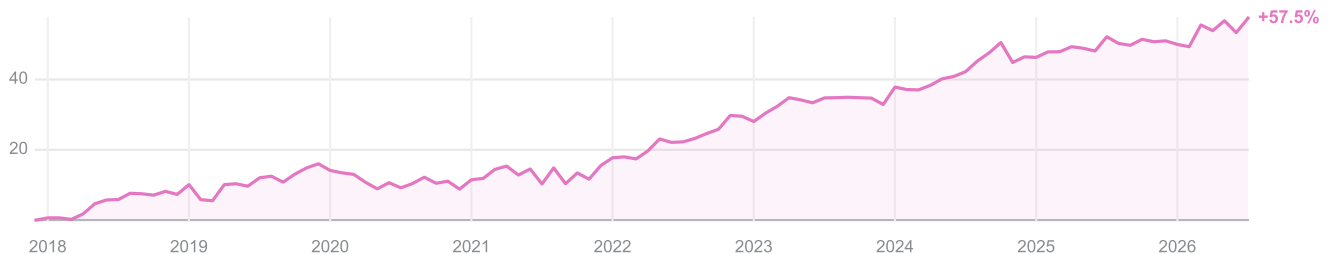
Price Momentum

CAGR +3.7%

036 ● **Adjusted Accruals**

Earnings Quality

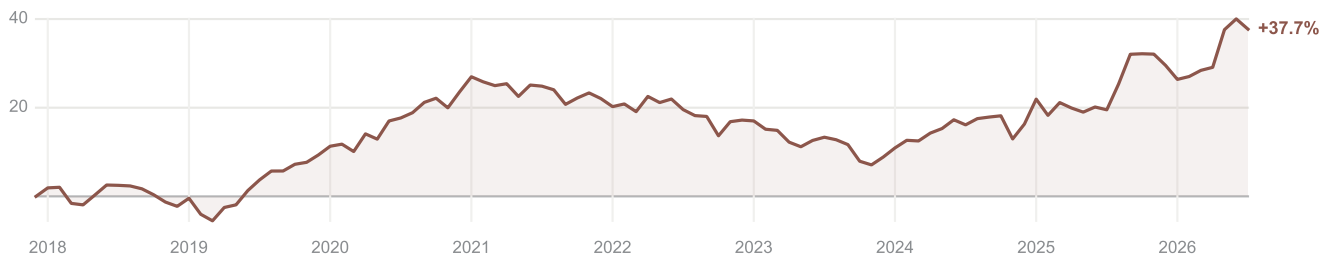
CAGR +3.7%



037 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth

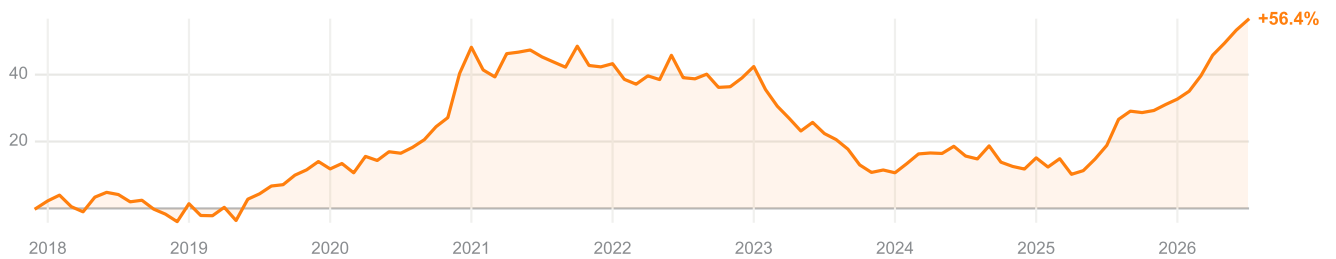
CAGR +3.6%



038 ● Institution Ownership Concentration

Price Momentum

CAGR +3.6%



039 ● Target Price Gap to 6M EMA

Analyst Expectations

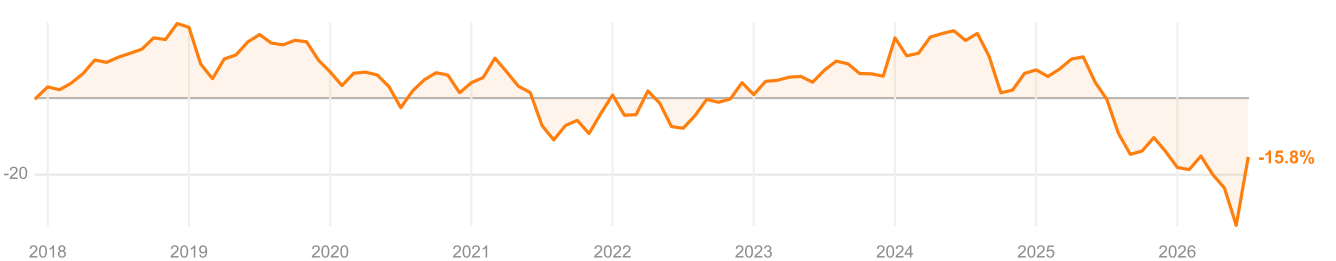
CAGR +3.5%



040 ● Share Turnover

Price Momentum

CAGR +3.5%



041 ● 1Y Chg in Sales to Price

Historical Growth

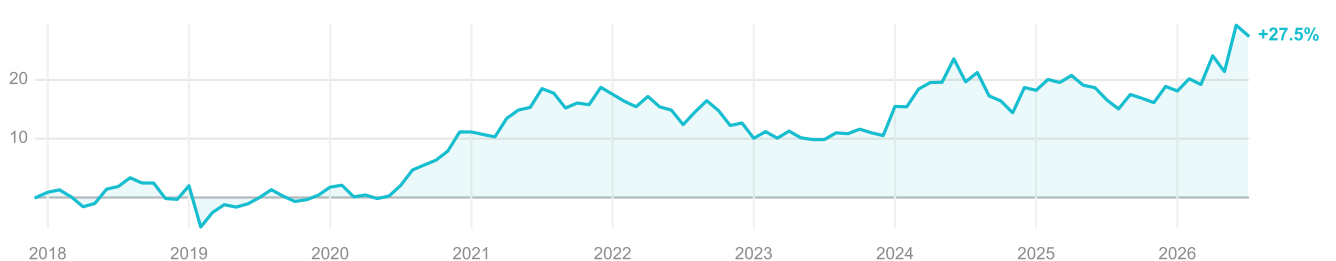
CAGR +3.5%



042 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

CAGR +3.4%



043 ● 1Y Chg in Asset Adjusted EPS

Historical Growth

CAGR +3.3%



044 ● 3Y Chg in Free Cash Flow per Share

Historical Growth

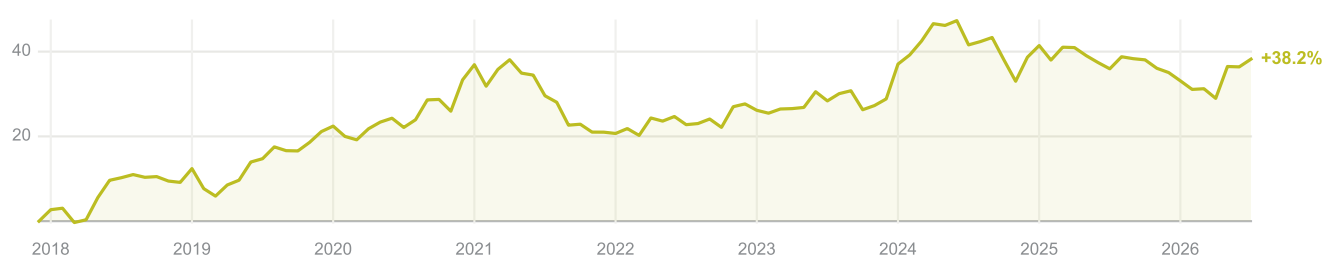
CAGR +3.3%



045 ● Capital Acquisition Ratio

Capital Efficiency

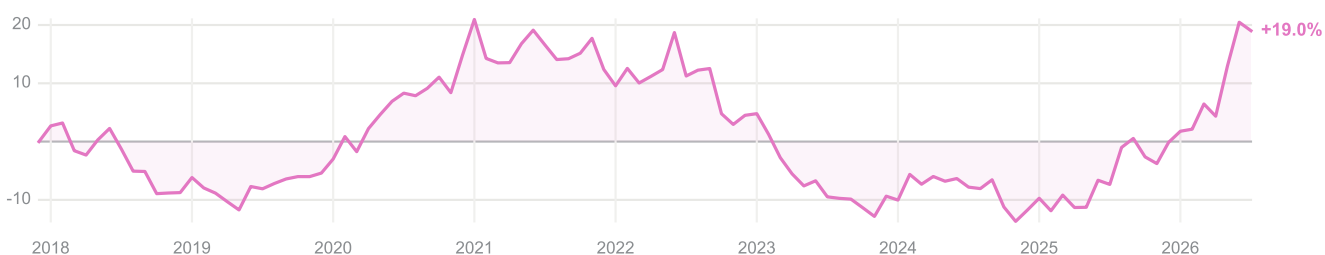
CAGR +3.2%



046 ● EBIT Margin

Earnings Quality

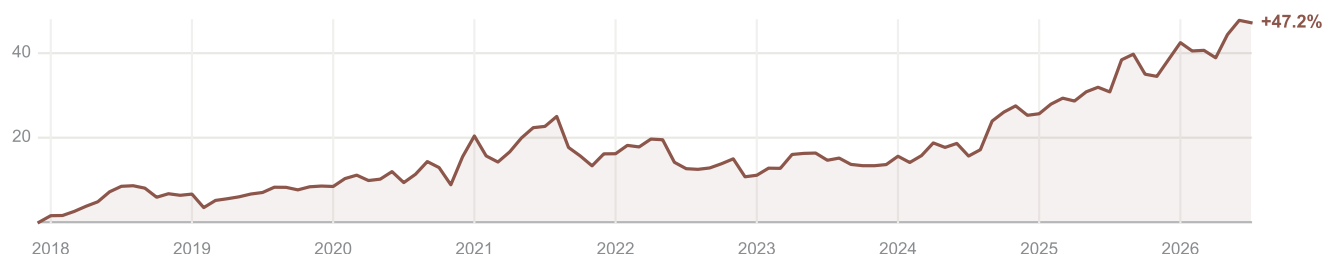
CAGR +3.1%



047 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth

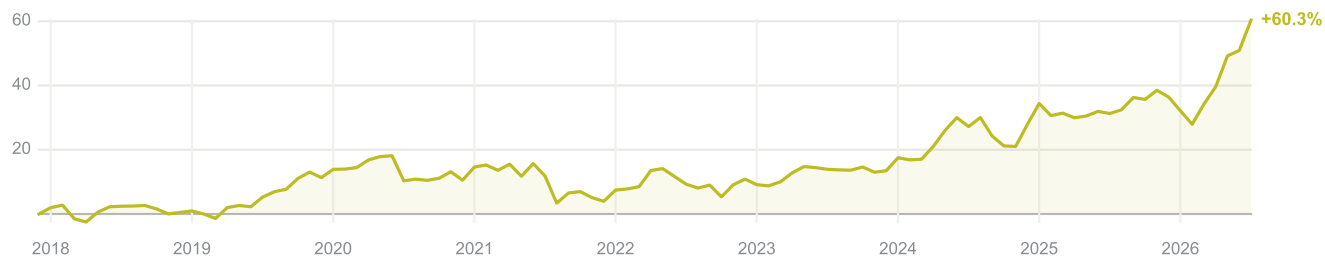
CAGR +3.1%



048 ● Receivables Turnover Ratio

Capital Efficiency

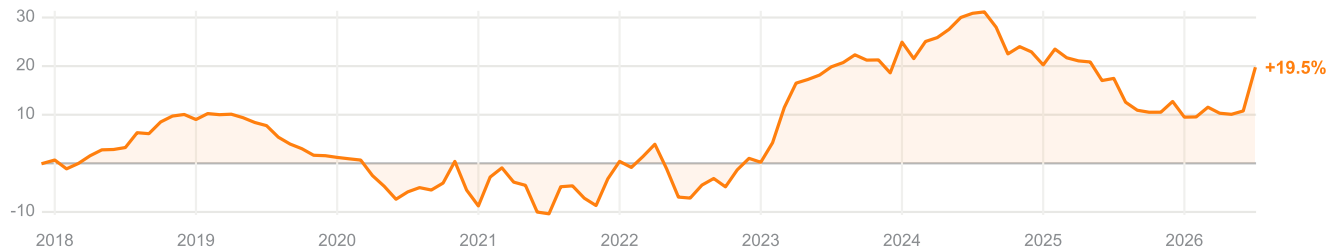
CAGR +3.1%



049 ● **Ownership Breadth Stability of Institution Ownership by Holders**

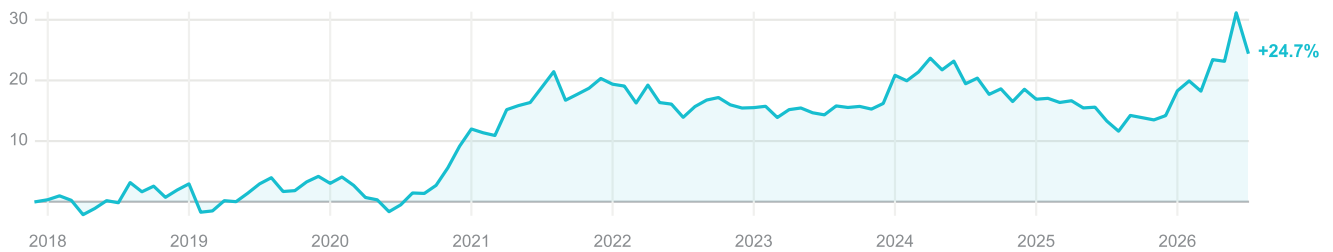
Price Momentum

CAGR +3.1%

050 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**

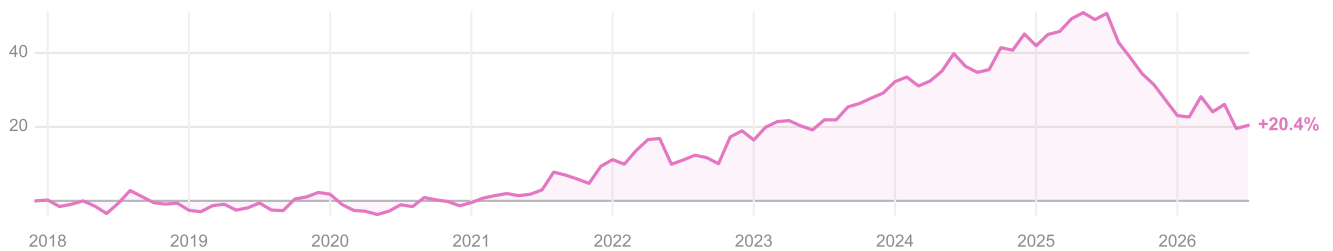
Analyst Expectations

CAGR +3.1%

051 ● **Working Capital Accruals**

Earnings Quality

CAGR +3.1%

052 ● **Assets to Price Ratio**

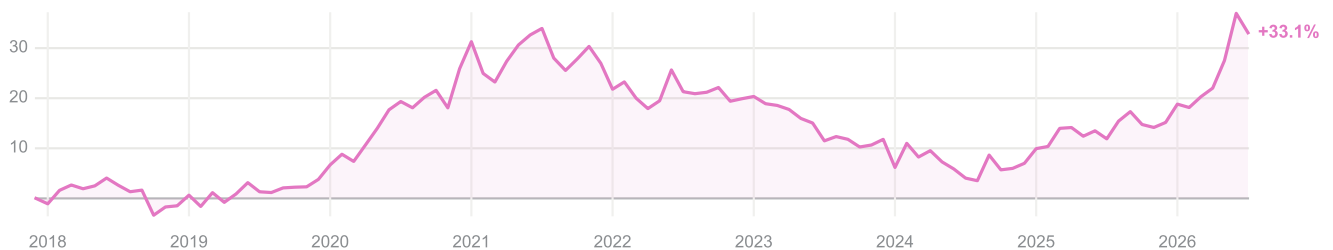
Valuation

CAGR +3.0%

053 ● **Gross Profit Margin**

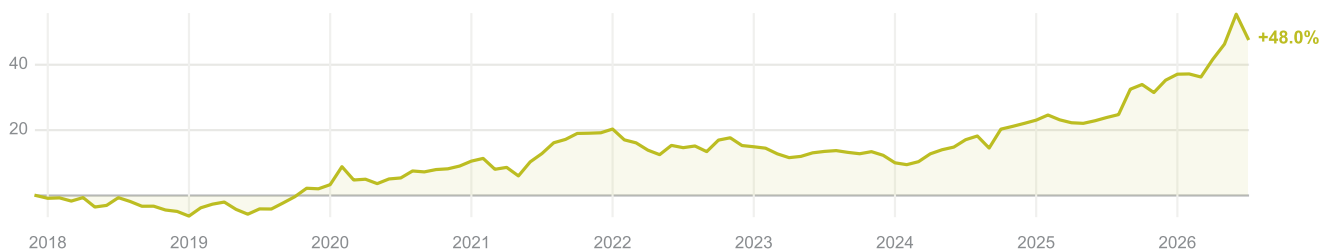
Earnings Quality

CAGR +3.0%

054 ● **Adjusted Interest Coverage Ratio**

Capital Efficiency

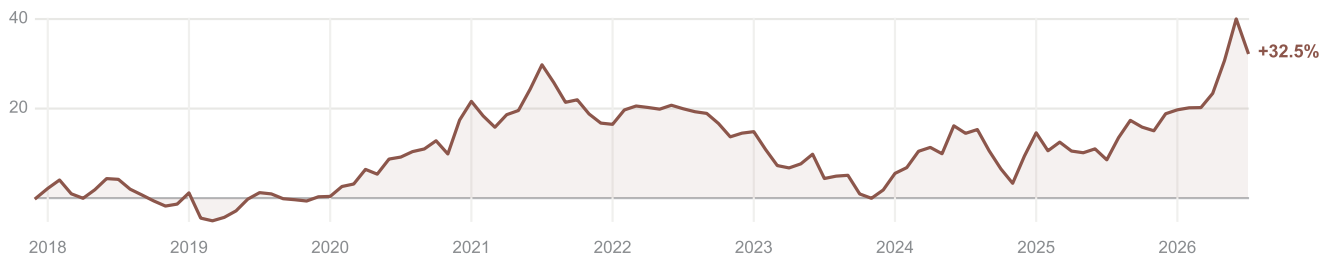
CAGR +2.9%



055 ● 1Y Chg in Cash Flow per Share

Historical Growth

CAGR +2.9%



056 ● 1Y Chg in Free Cash Flow per Share

Historical Growth

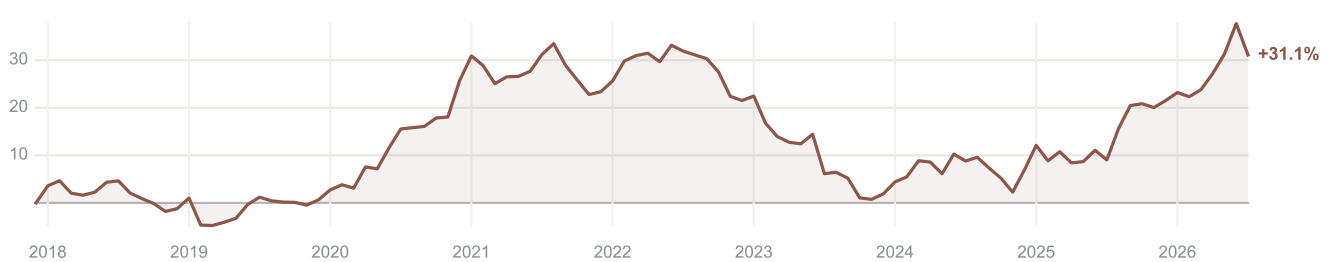
CAGR +2.9%



057 ● 1Y Chg in Cash Flow to Price

Historical Growth

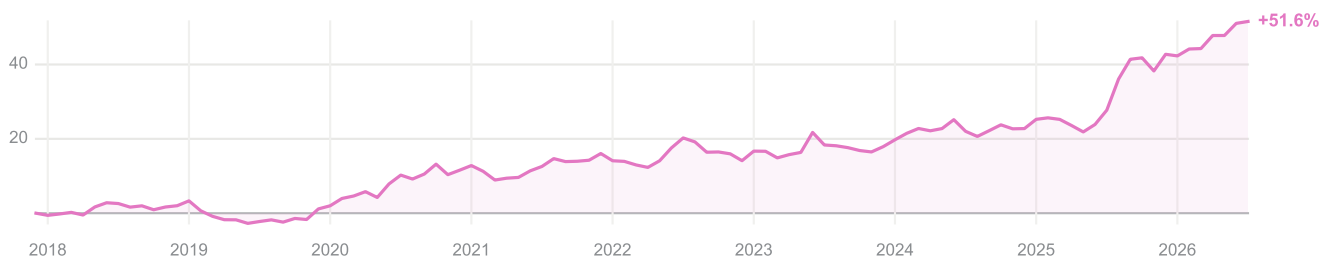
CAGR +2.9%



058 ● Sales to Invested Capital

Earnings Quality

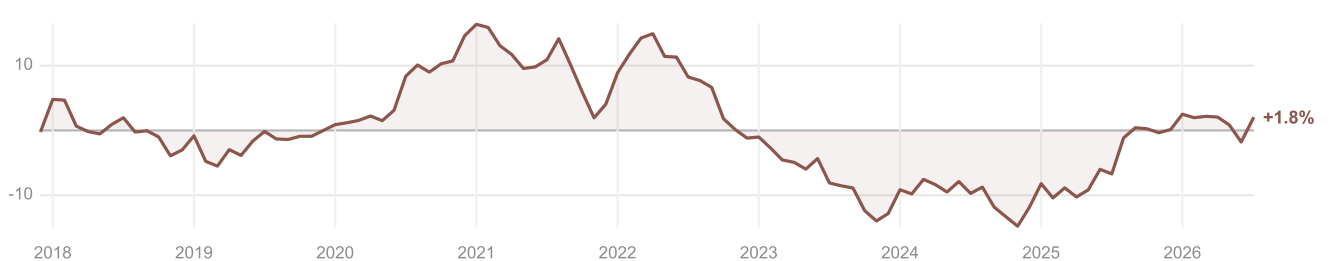
CAGR +2.9%



059 ● 1Y Chg in Earnings to Price

Historical Growth

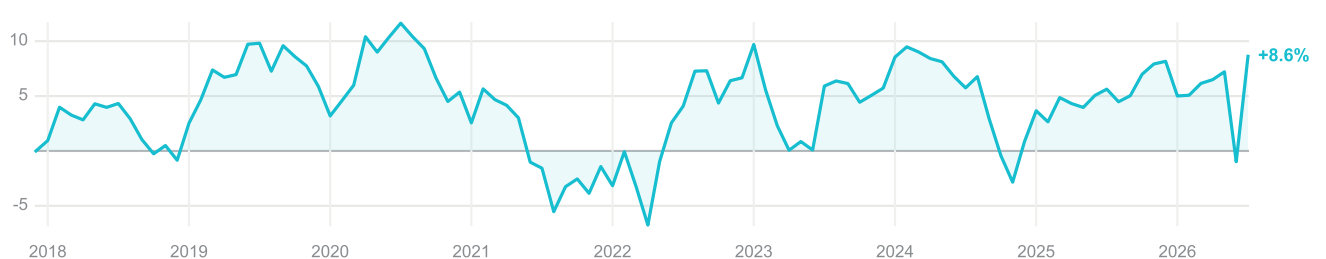
CAGR +2.9%



060 ● 6M Chg in Target Price Gap

Analyst Expectations

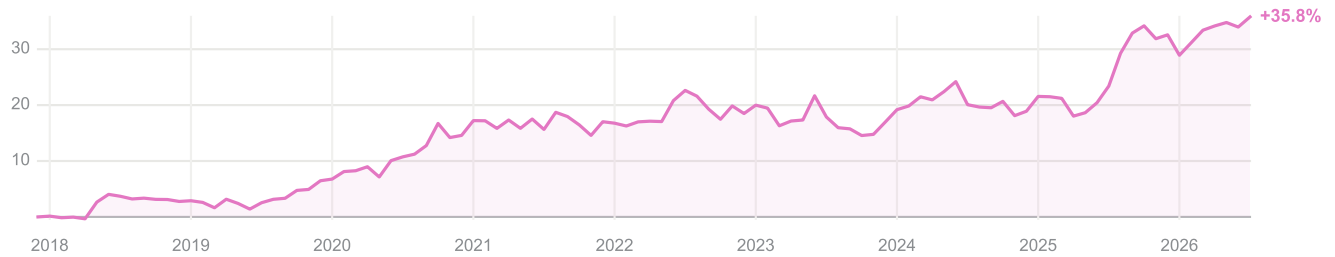
CAGR +2.9%



061 ● Equity Turnover

Earnings Quality

CAGR +2.8%

**062 ● Net Cash Flow to Price**

Valuation

CAGR +2.8%

**063 ● Cash to Price**

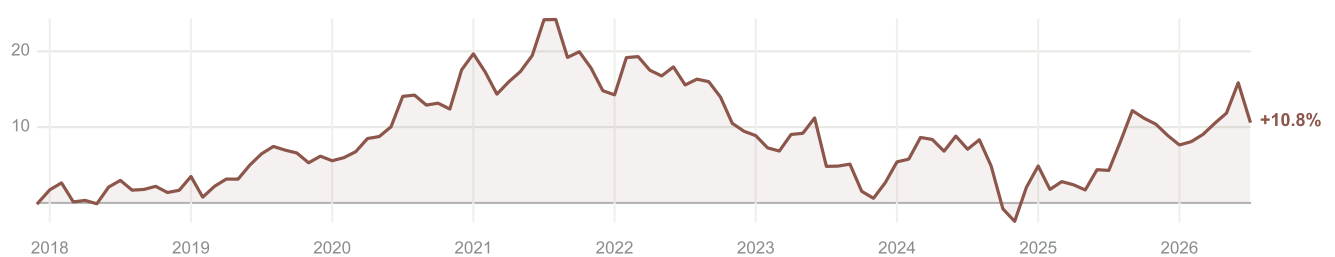
Valuation

CAGR +2.7%

**064 ● 1Y Chg in Operating Margin**

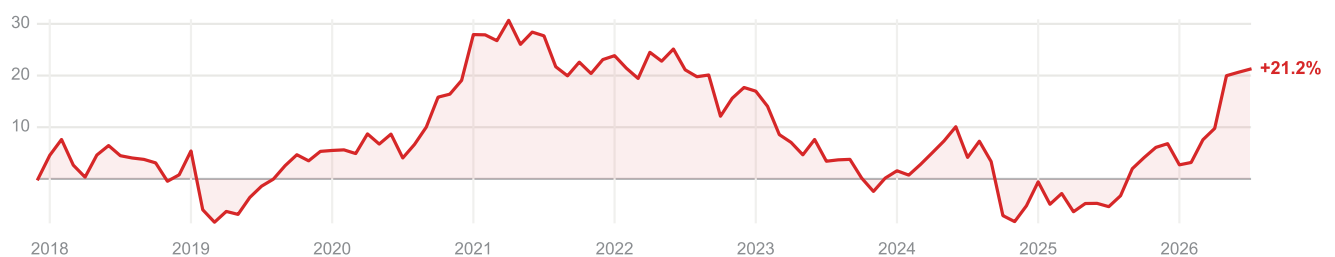
Historical Growth

CAGR +2.7%

**065 ● Cash Flow to Equity**

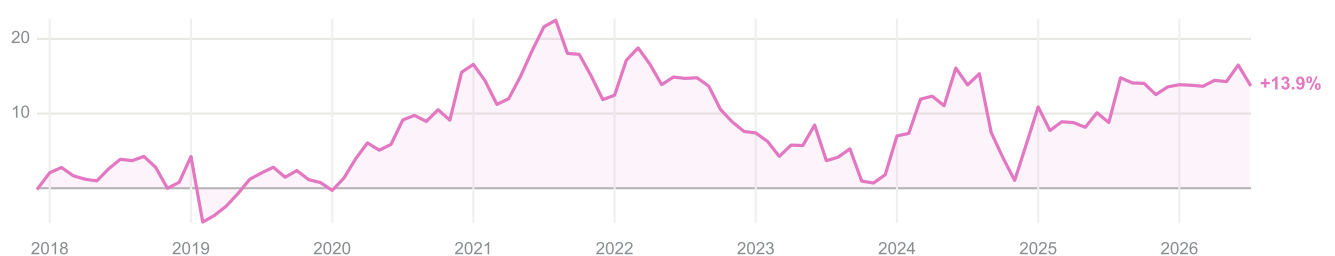
Valuation

CAGR +2.7%

**066 ● Change EPS to Sales**

Earnings Quality

CAGR +2.7%



067 ● Growth Flow to Price

Valuation

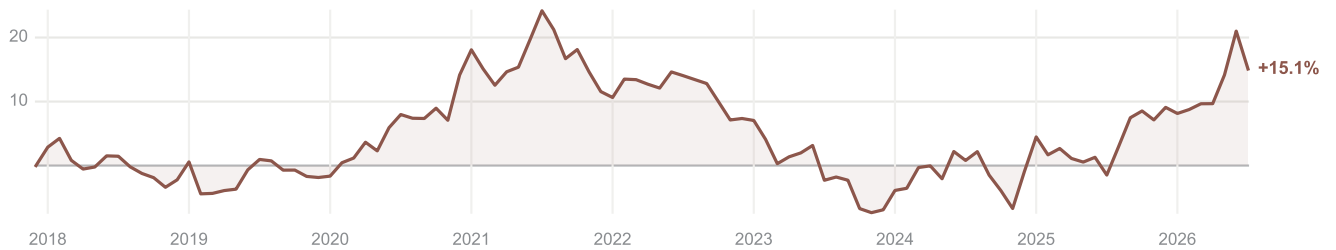
CAGR +2.7%



068 ● 1Y Chg in Asset Adjusted Cash Flow

Historical Growth

CAGR +2.7%



069 ● 1Y Chg in Earnings per Share

Historical Growth

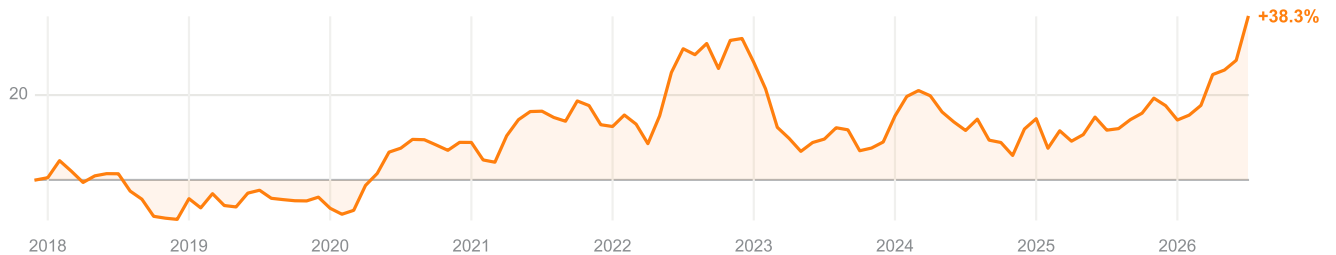
CAGR +2.6%



070 ● Risk Adjusted Relative Strength

Price Momentum

CAGR +2.6%



071 ● Free Cash Flow to Price

Valuation

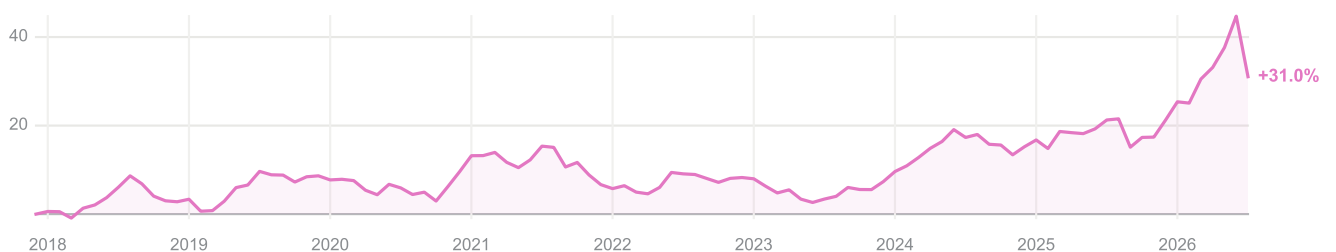
CAGR +2.6%



072 ● Unexpected Receivables Change

Earnings Quality

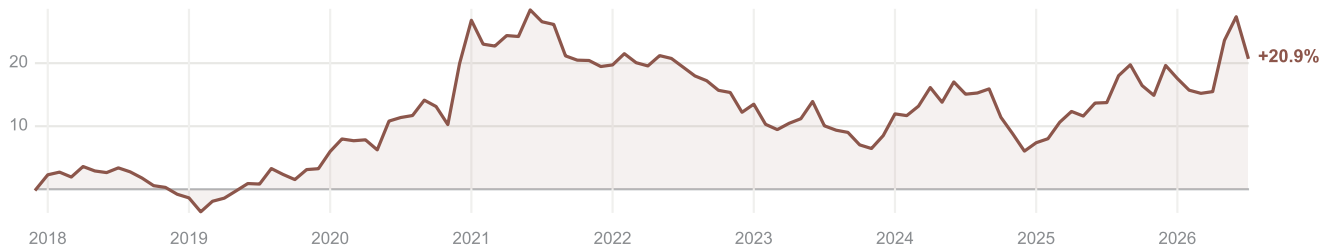
CAGR +2.6%



073 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +2.6%



074 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth

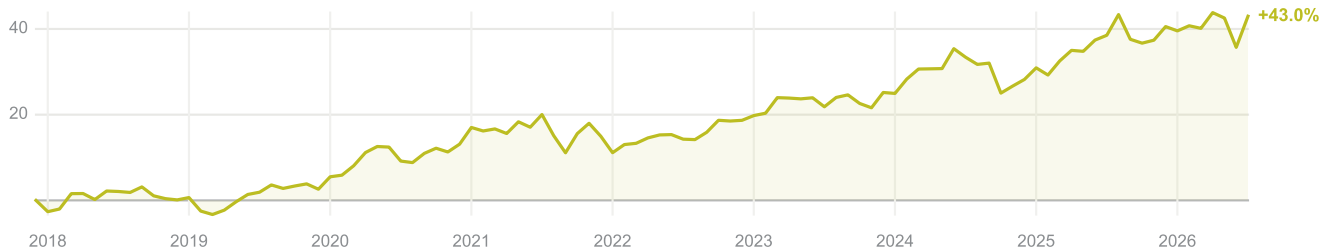
CAGR +2.5%



075 ● Long Term Debt to Assets Ratio

Capital Efficiency

CAGR +2.5%



076 ● 3Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +2.5%



077 ● Adjusted EBIT to Price

Valuation

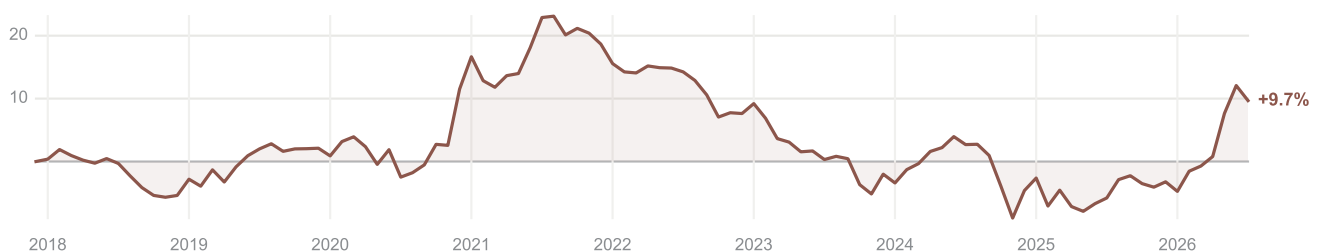
CAGR +2.4%

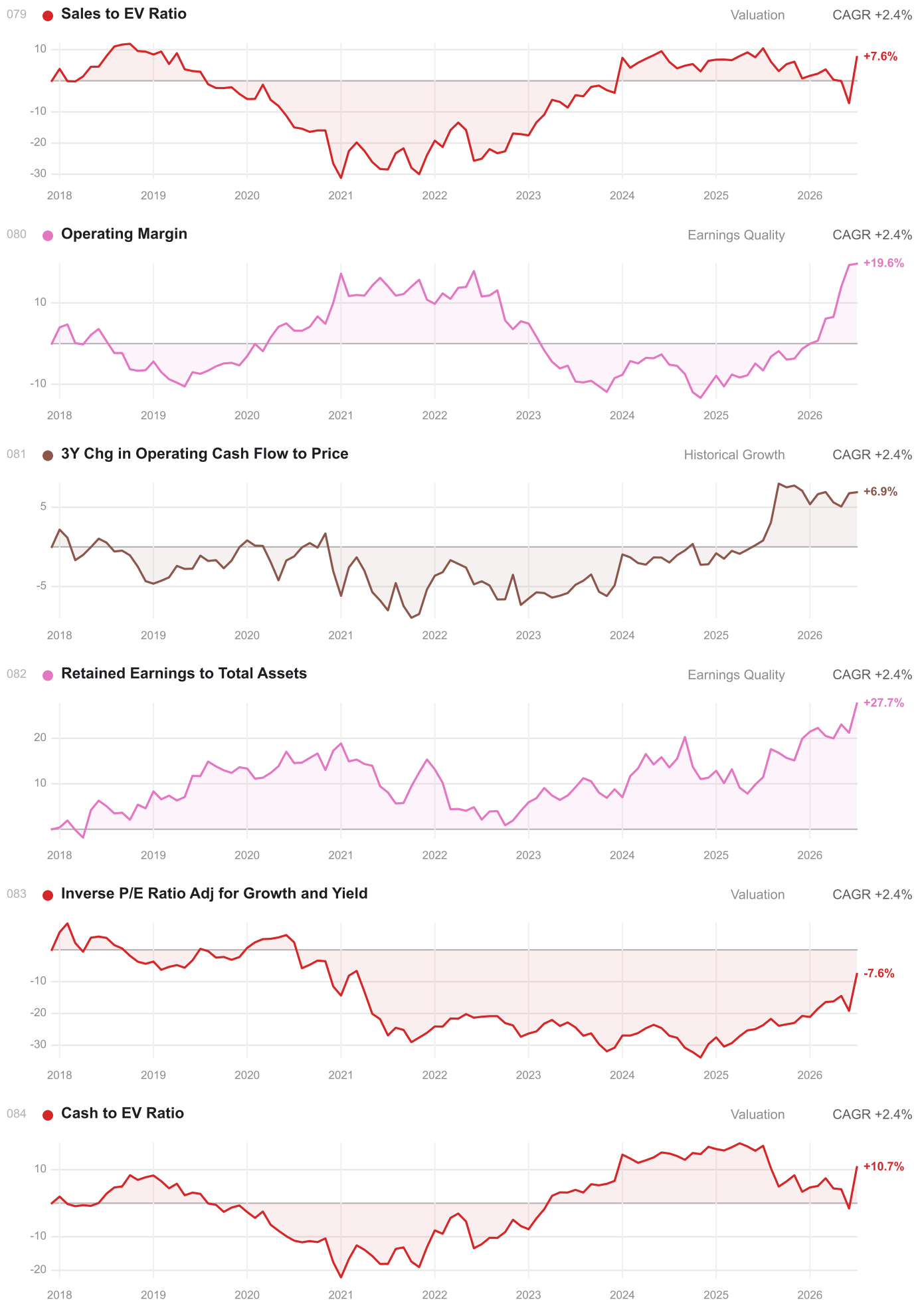


078 ● 3Y Average Annual Earnings Growth

Historical Growth

CAGR +2.4%





085 ● **Net Cash Flow to Enterprise Value**

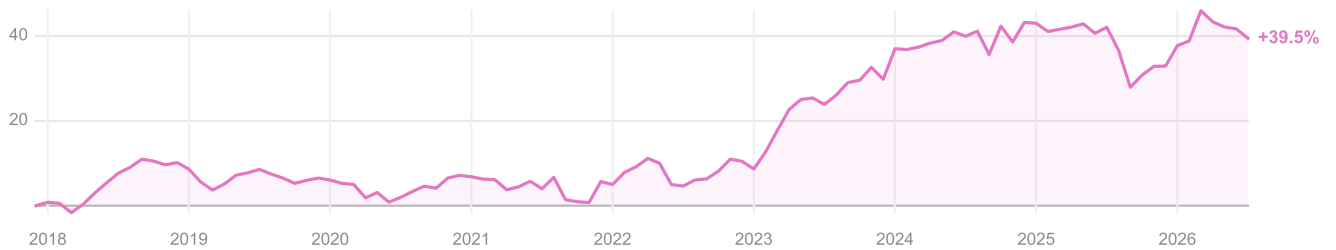
Valuation

CAGR +2.4%

086 ● **Accrual Ratio - Balance Sheet**

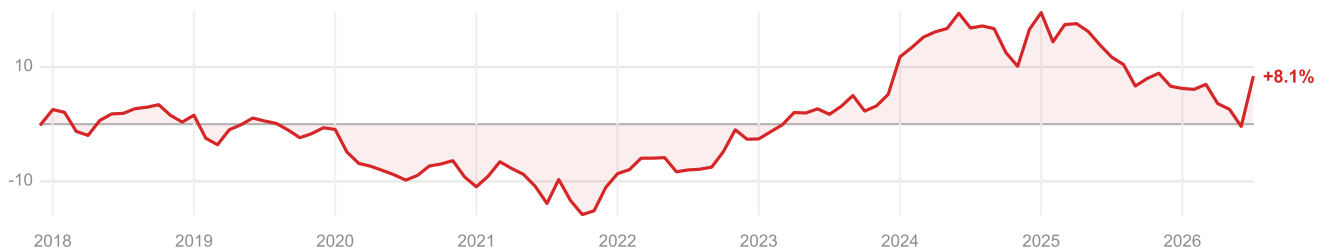
Earnings Quality

CAGR +2.4%

087 ● **Free Cash Flow to Enterprise Value**

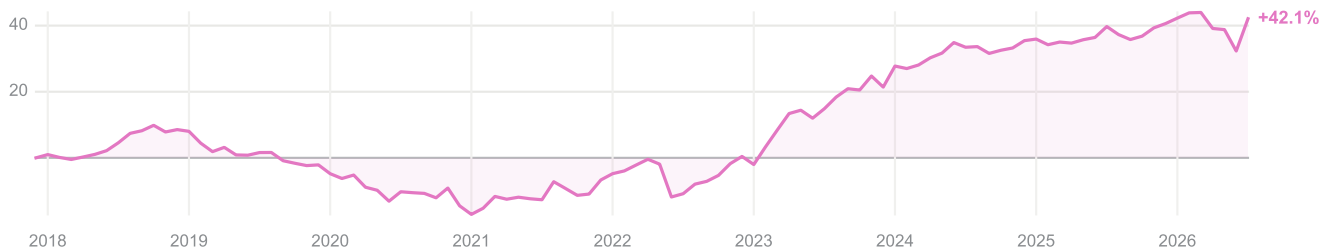
Valuation

CAGR +2.3%

088 ● **Accrual Ratio - Cash Flows**

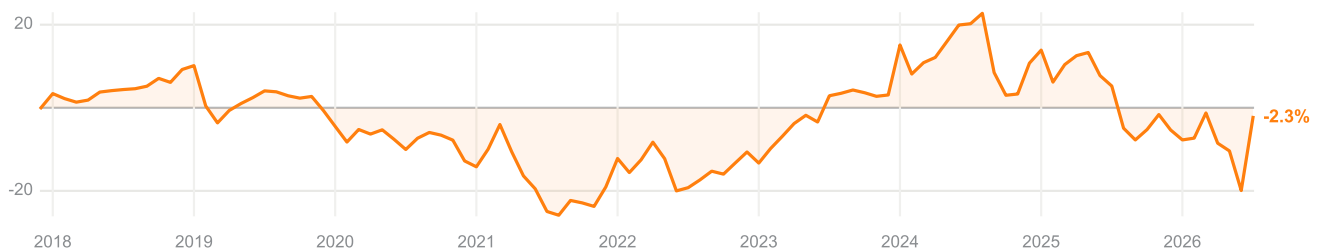
Earnings Quality

CAGR +2.3%

089 ● **Max Daily Return in the Past 6 Months**

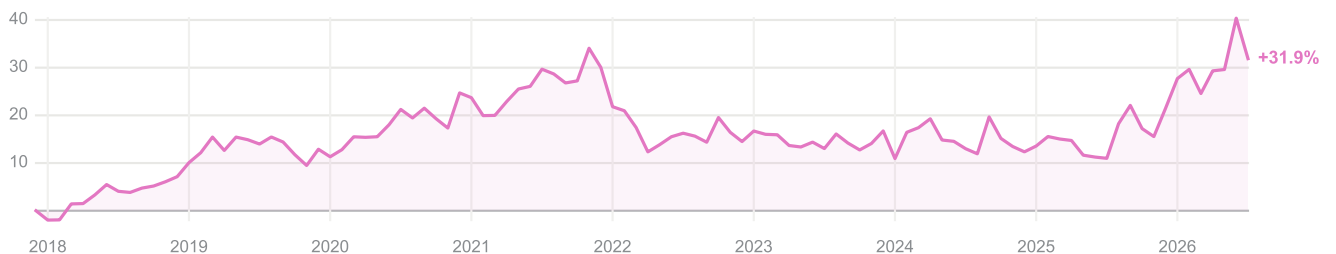
Price Momentum

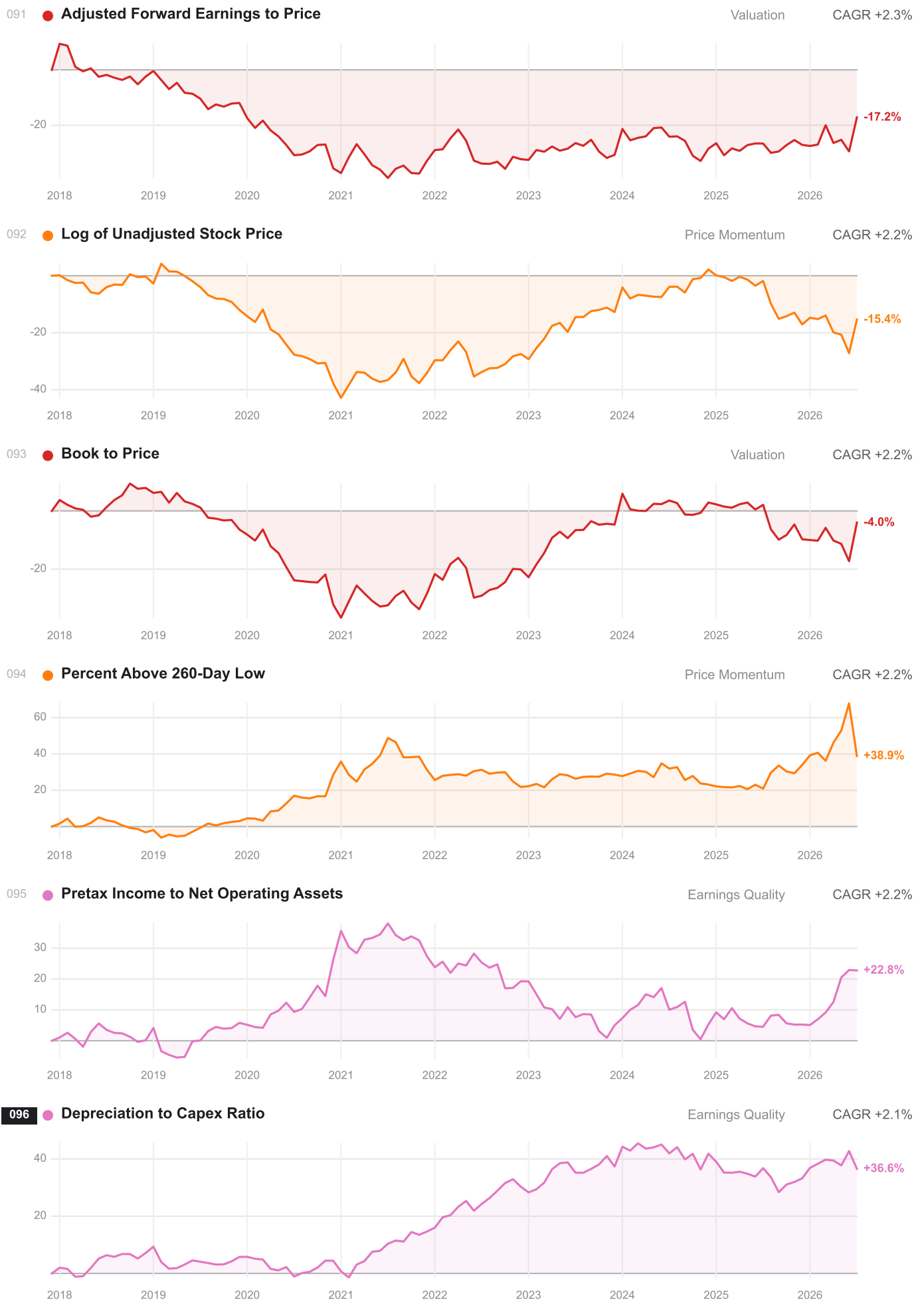
CAGR +2.3%

090 ● **Working Capital to Assets Ratio**

Earnings Quality

CAGR +2.3%

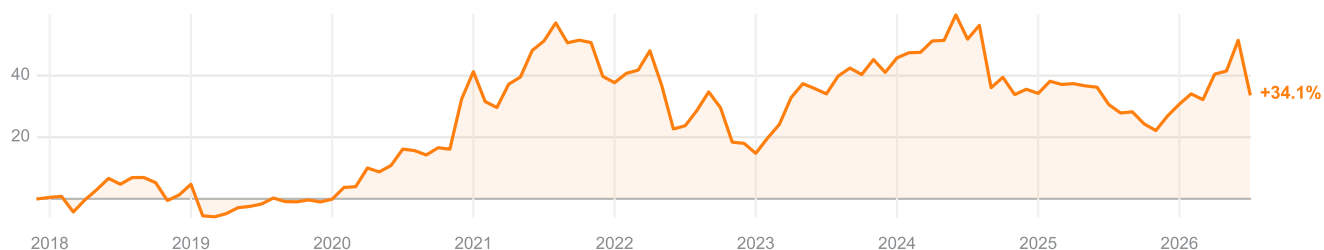




097 ● 15/36 Week Price Ratio

Price Momentum

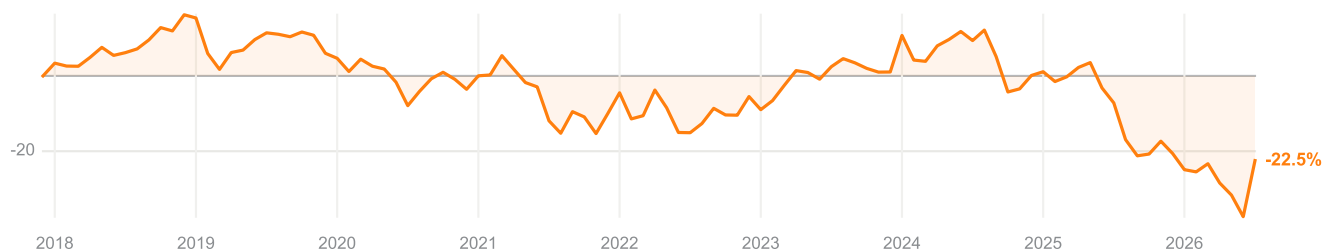
CAGR +2.1%



098 ● Adjusted 6 Month Average Share Turn Over

Price Momentum

CAGR +2.1%



099 ● Solvency Ratio

Valuation

CAGR +2.1%



100 ● Cash to Total Assets

Valuation

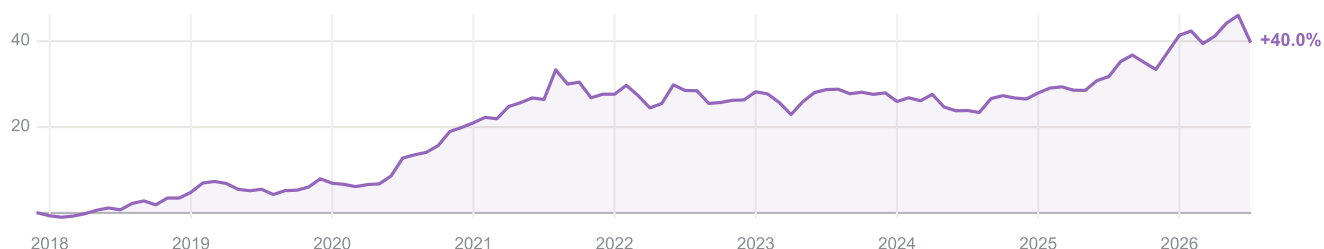
CAGR +2.1%



101 ● 60M CAPM Beta

Volatility

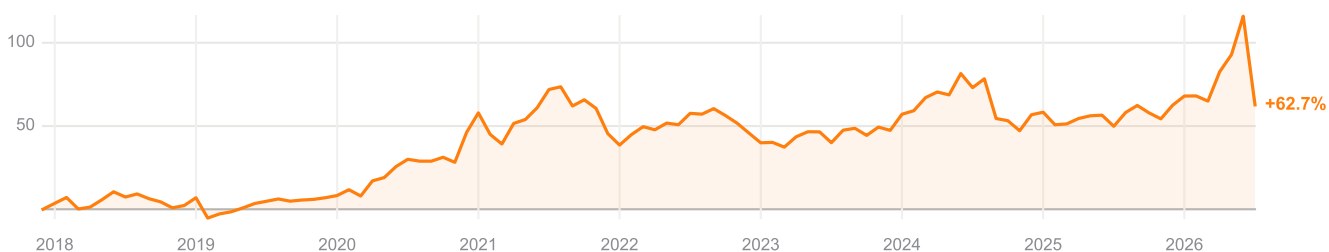
CAGR +2.1%

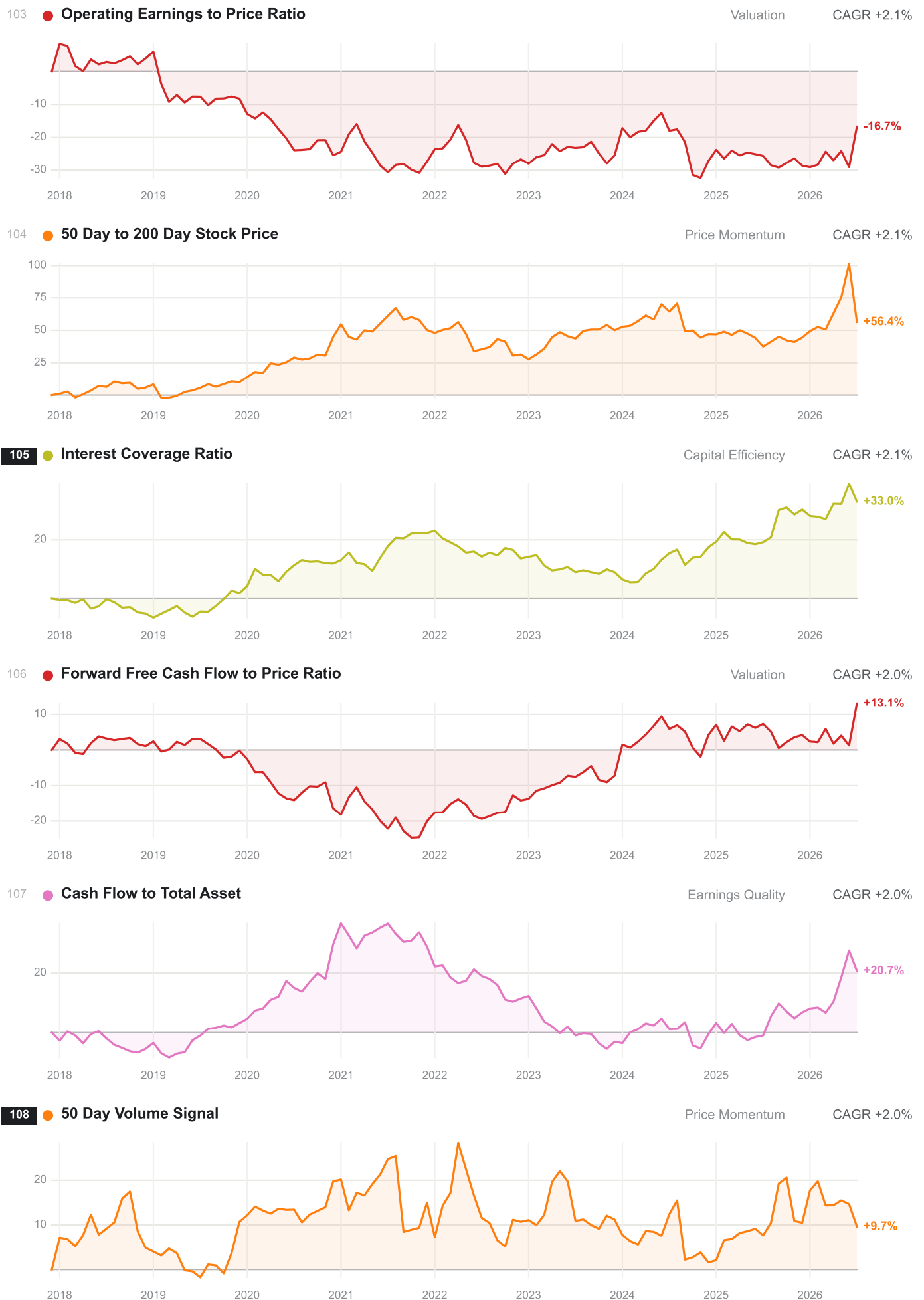


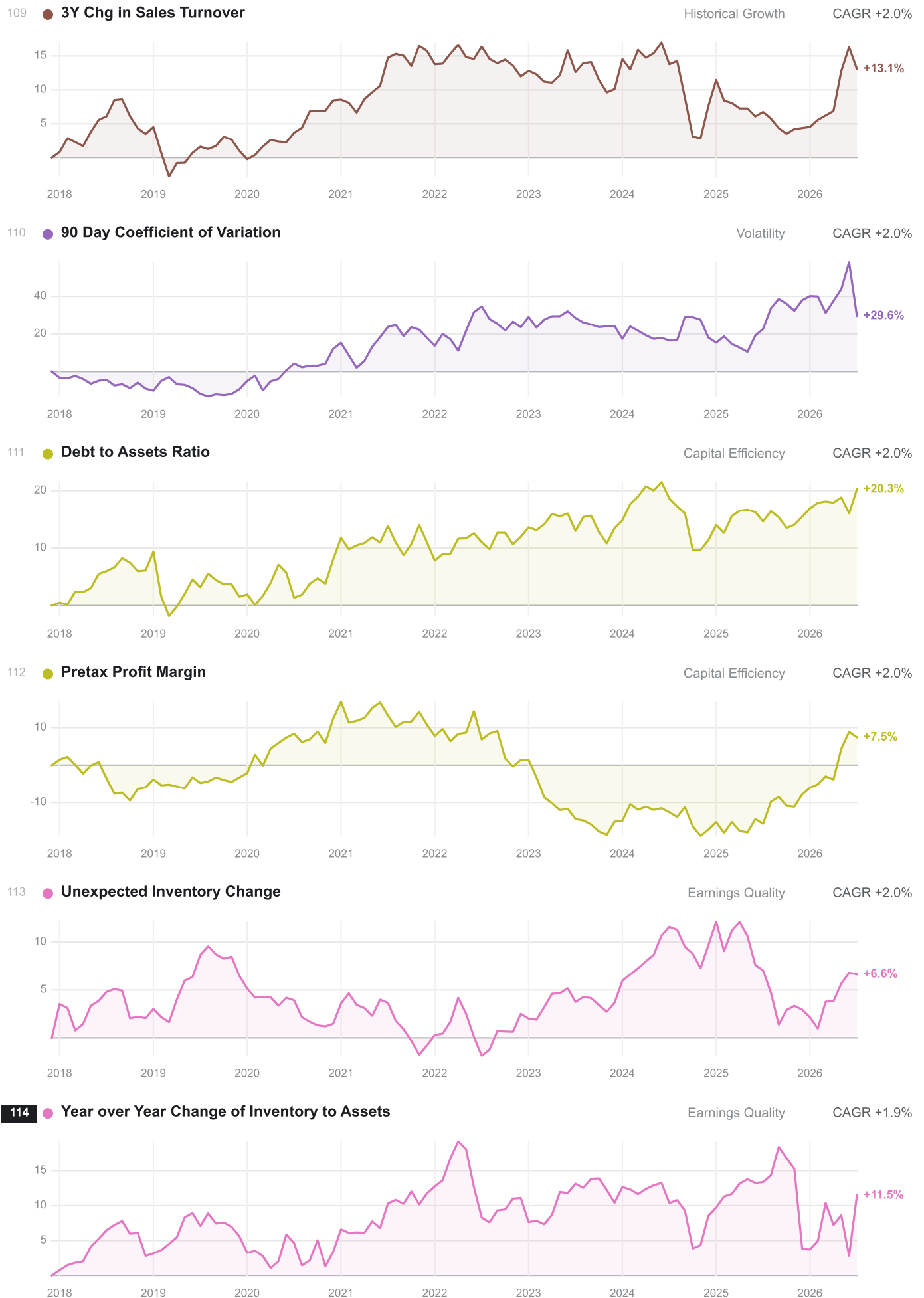
102 ● 12M - 1M Price Momentum

Price Momentum

CAGR +2.1%



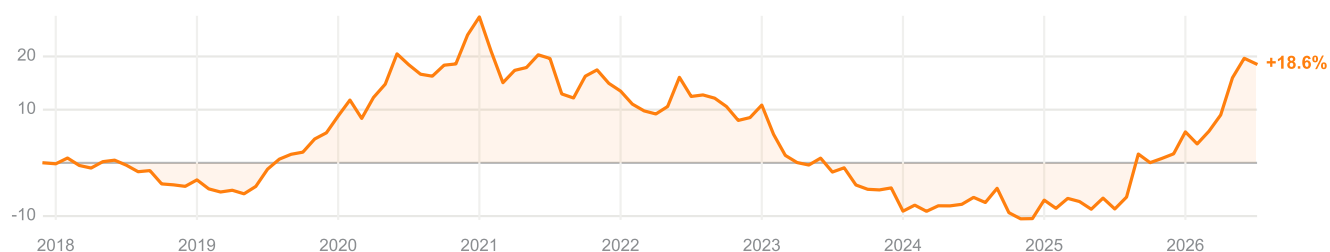




115 ● **Institution Ownership Level**

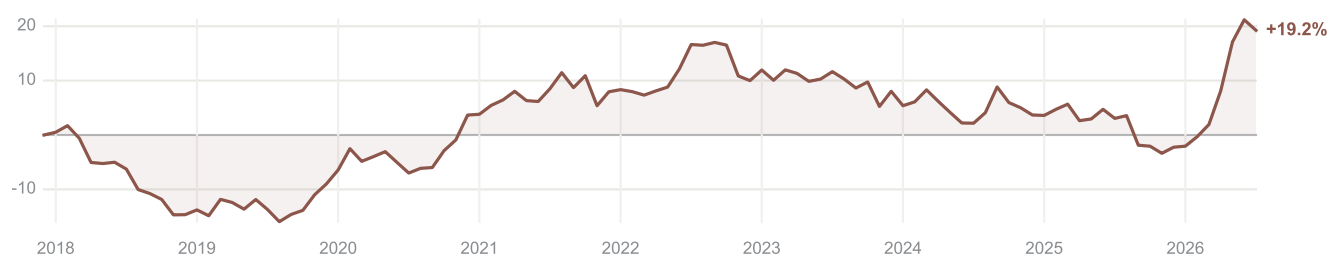
Price Momentum

CAGR +1.9%

116 ● **Dividend Growth**

Historical Growth

CAGR +1.9%

117 ● **Year over Year Change of EPS to Sales**

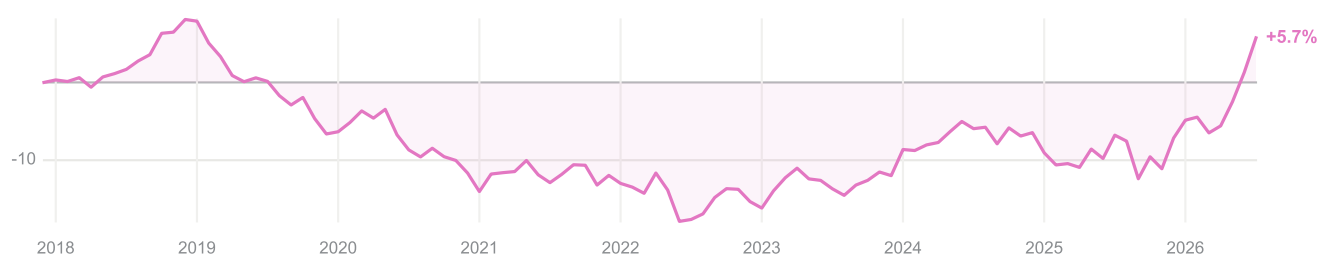
Earnings Quality

CAGR +1.9%

118 ● **Change in TTM Depr. to Capex**

Earnings Quality

CAGR +1.9%

119 ● **3Y Chg in Asset Adjusted EPS**

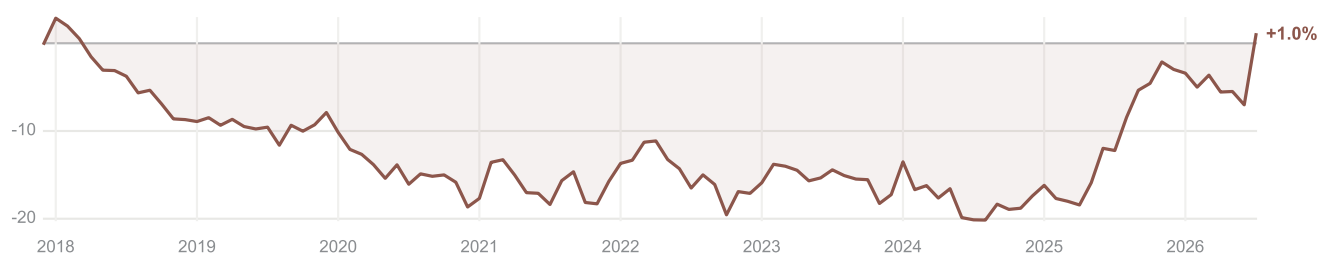
Historical Growth

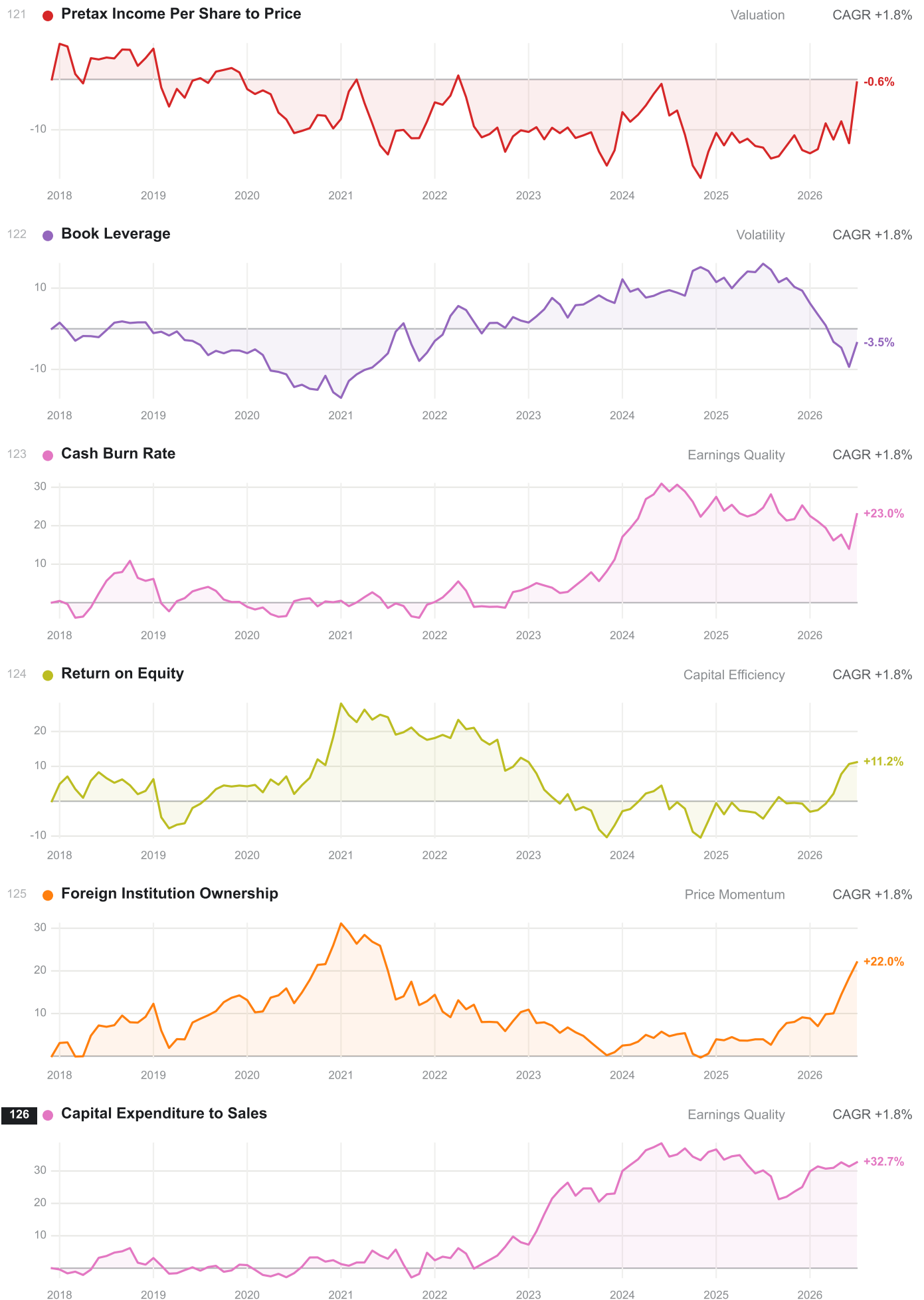
CAGR +1.8%

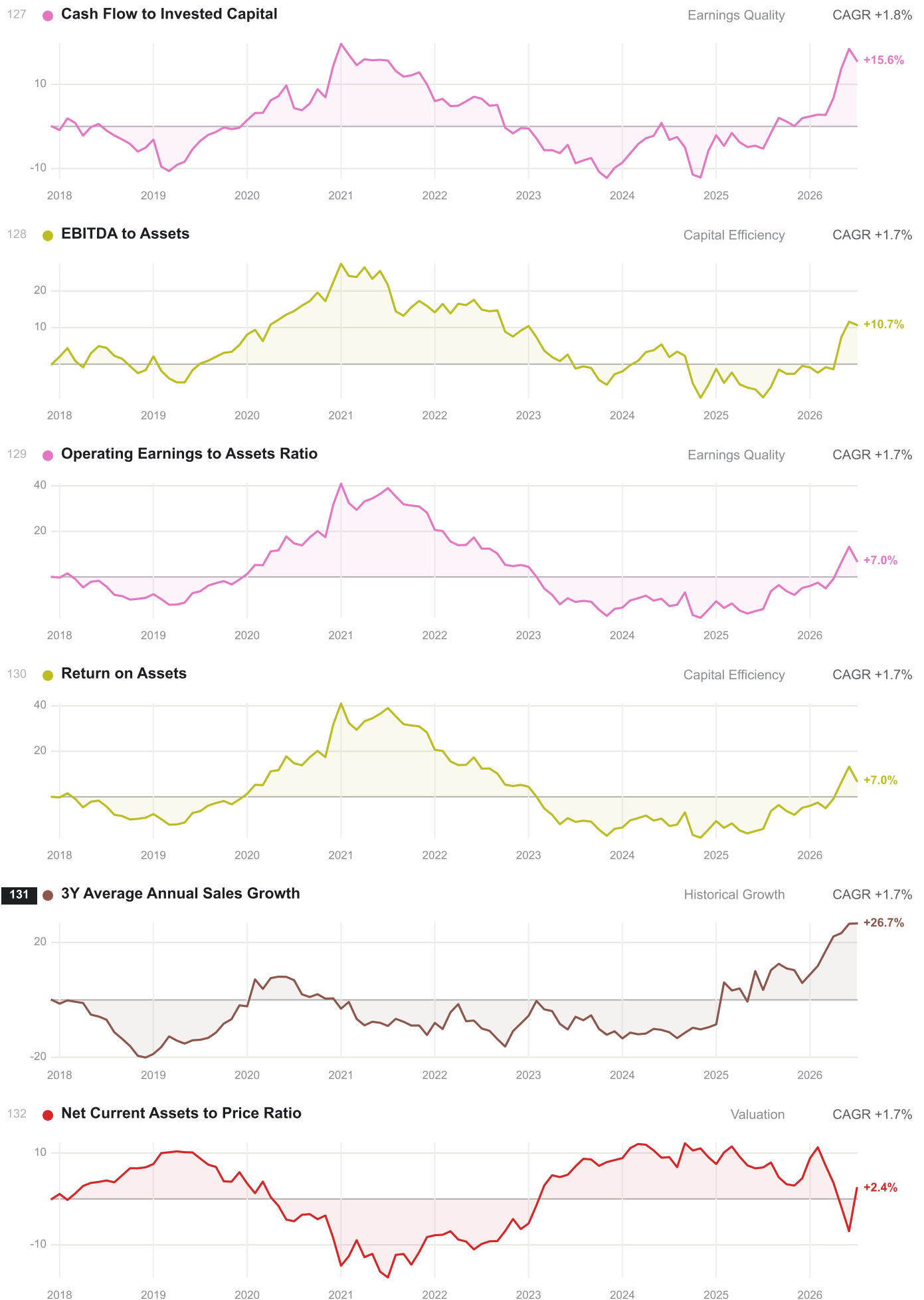
120 ● **3Y Chg in Cash Flow to Price**

Historical Growth

CAGR +1.8%



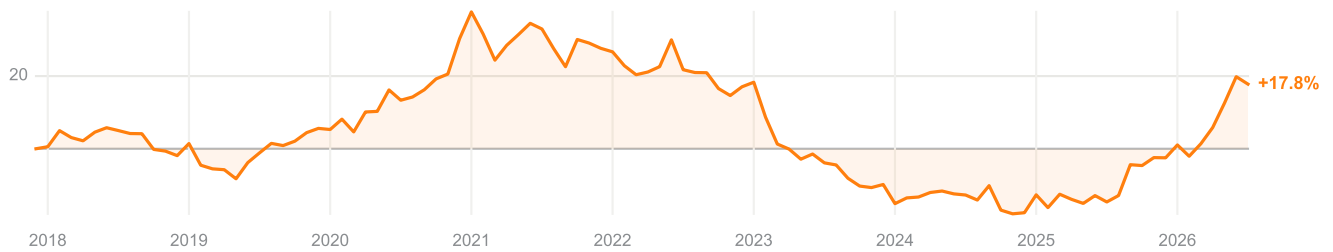




133 ● **Institution Ownership Level - Active Manager**

Price Momentum

CAGR +1.7%

134 ● **Log of Market Capitalization Cubed**

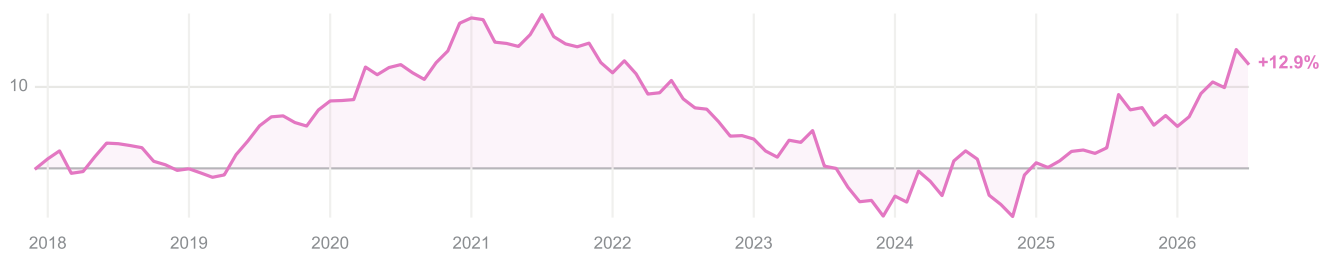
Size

CAGR +1.7%

135 ● **Year over Year Change in SGA to Sales**

Earnings Quality

CAGR +1.7%

136 ● **3Y Chg in Free Cash Flow to Price**

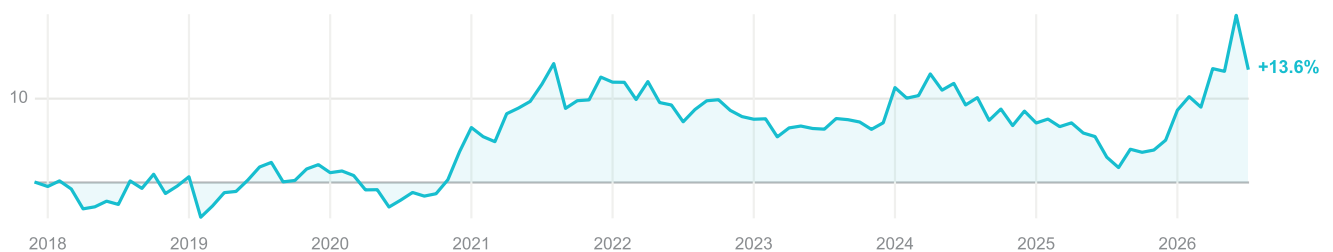
Historical Growth

CAGR +1.7%

137 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

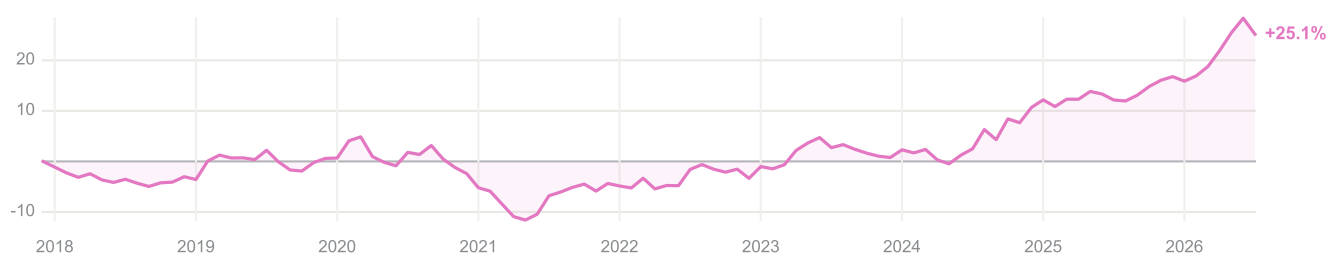
Analyst Expectations

CAGR +1.6%

138 ● **SG&A to Sales**

Earnings Quality

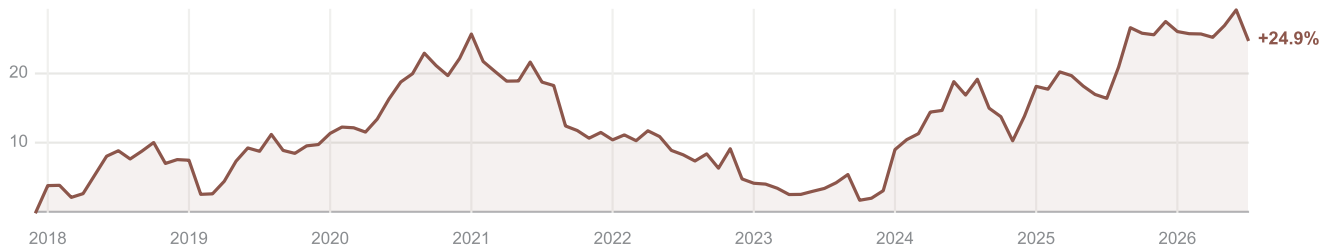
CAGR +1.6%



139 ● **3Y Chg in Asset Adjusted Free Cash Flow**

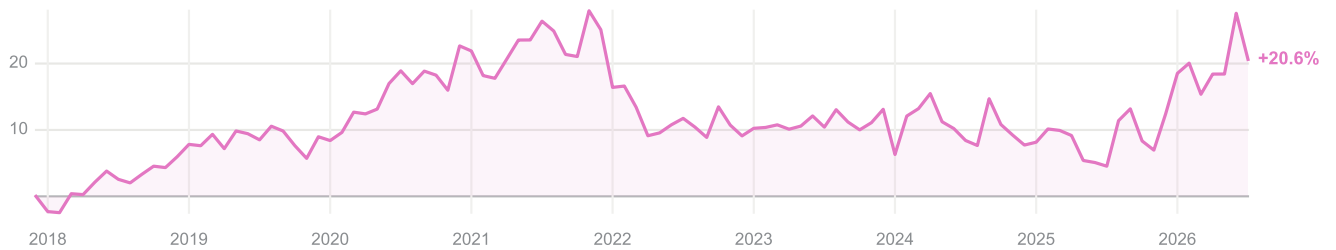
Historical Growth

CAGR +1.6%

140 ● **Working Capital to Total Assets**

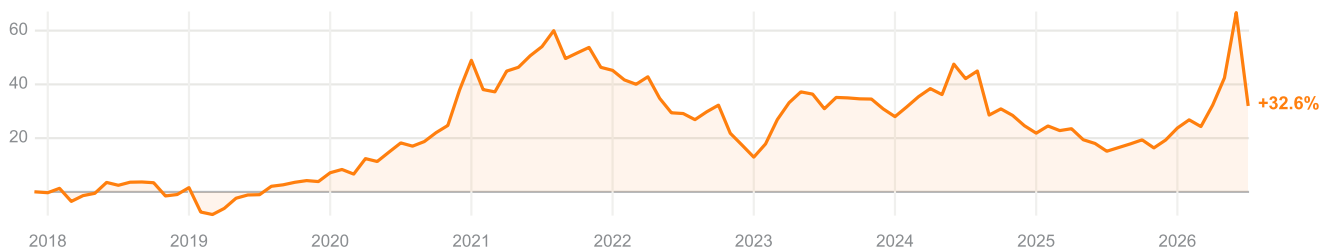
Earnings Quality

CAGR +1.6%

141 ● **26W Price Stat**

Price Momentum

CAGR +1.6%

142 ● **Dividends to Price Ratio**

Valuation

CAGR +1.5%

143 ● **Return on Net Tangible Assets**

Valuation

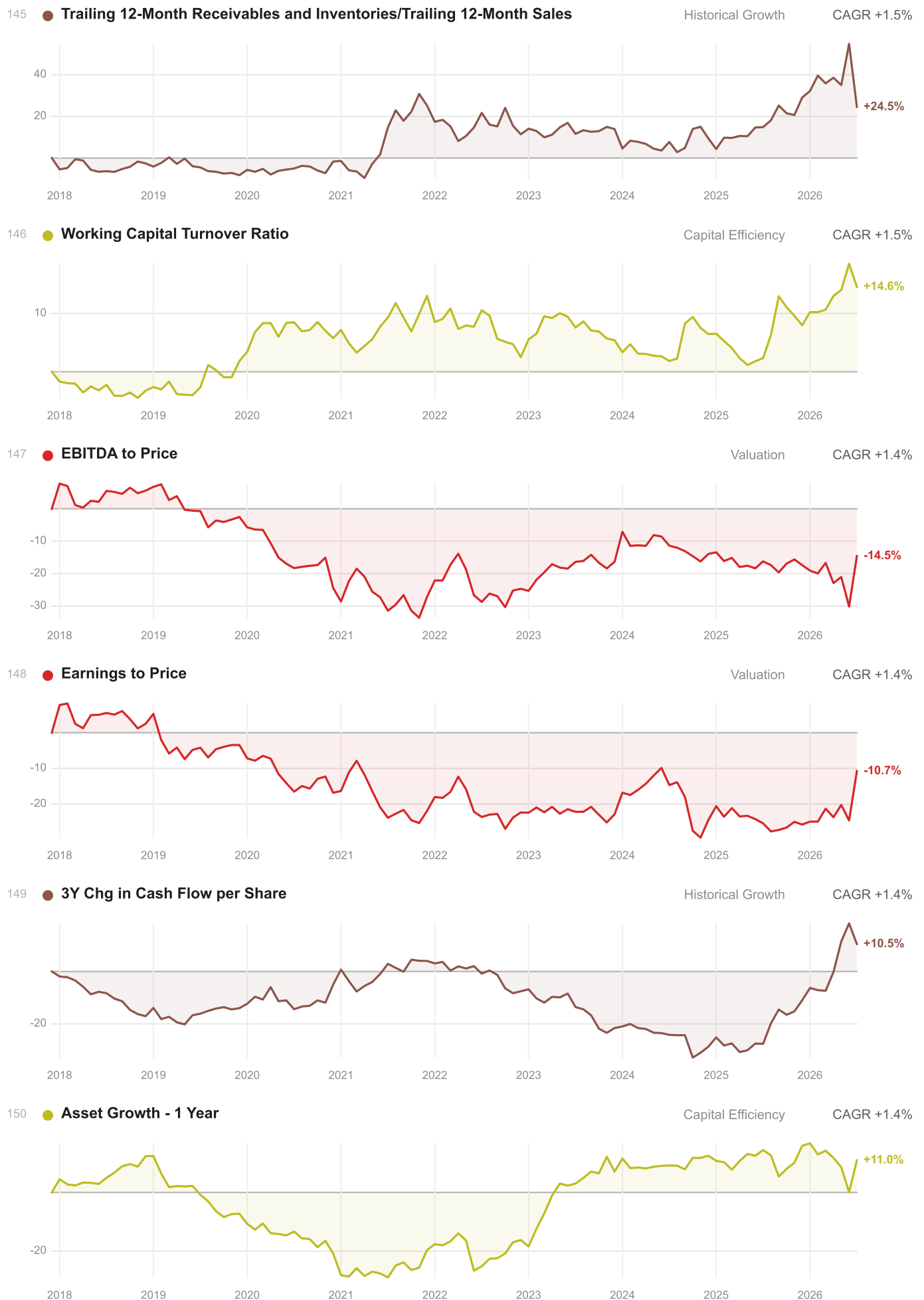
CAGR +1.5%

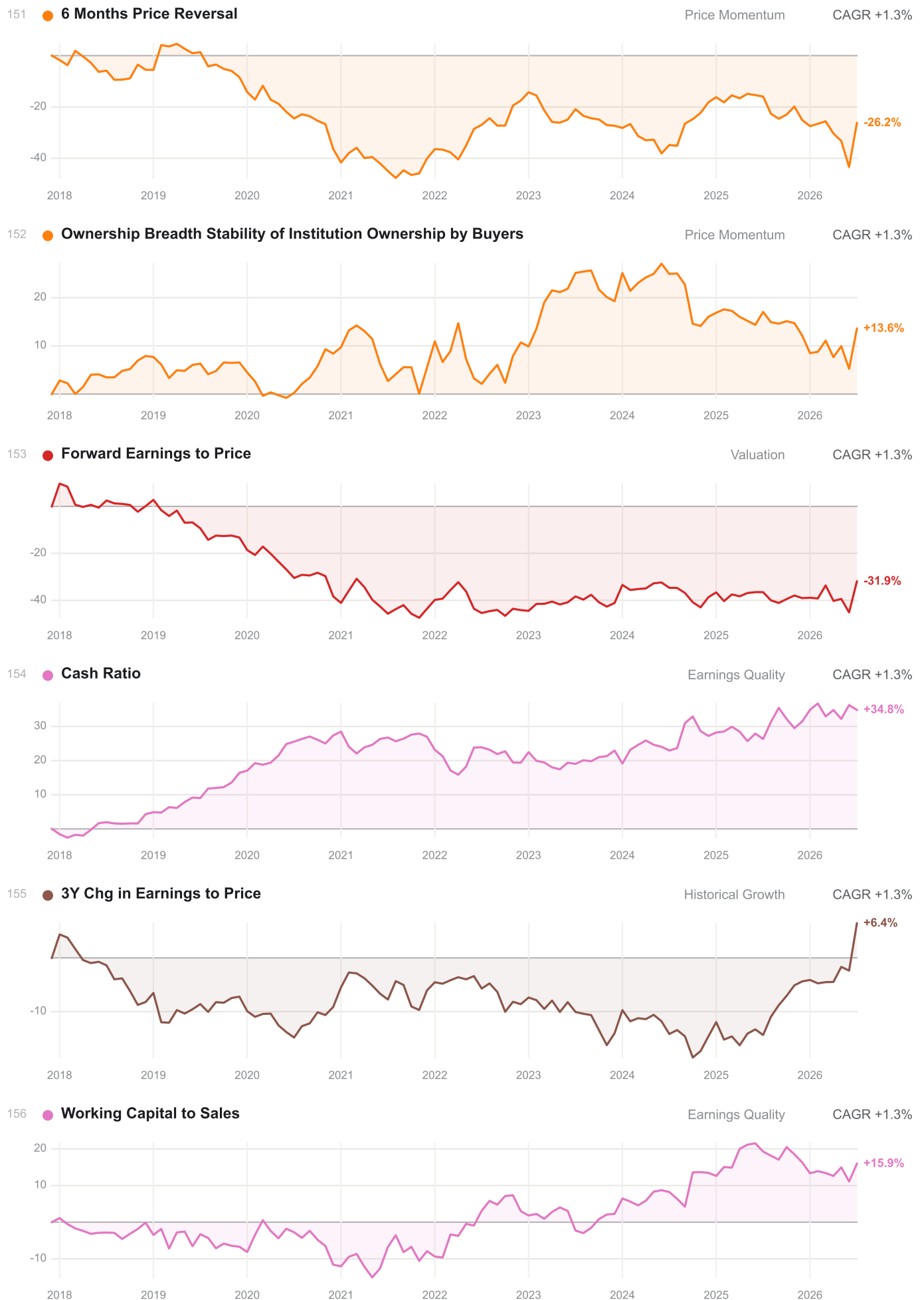
144 ● **Log Market Cap**

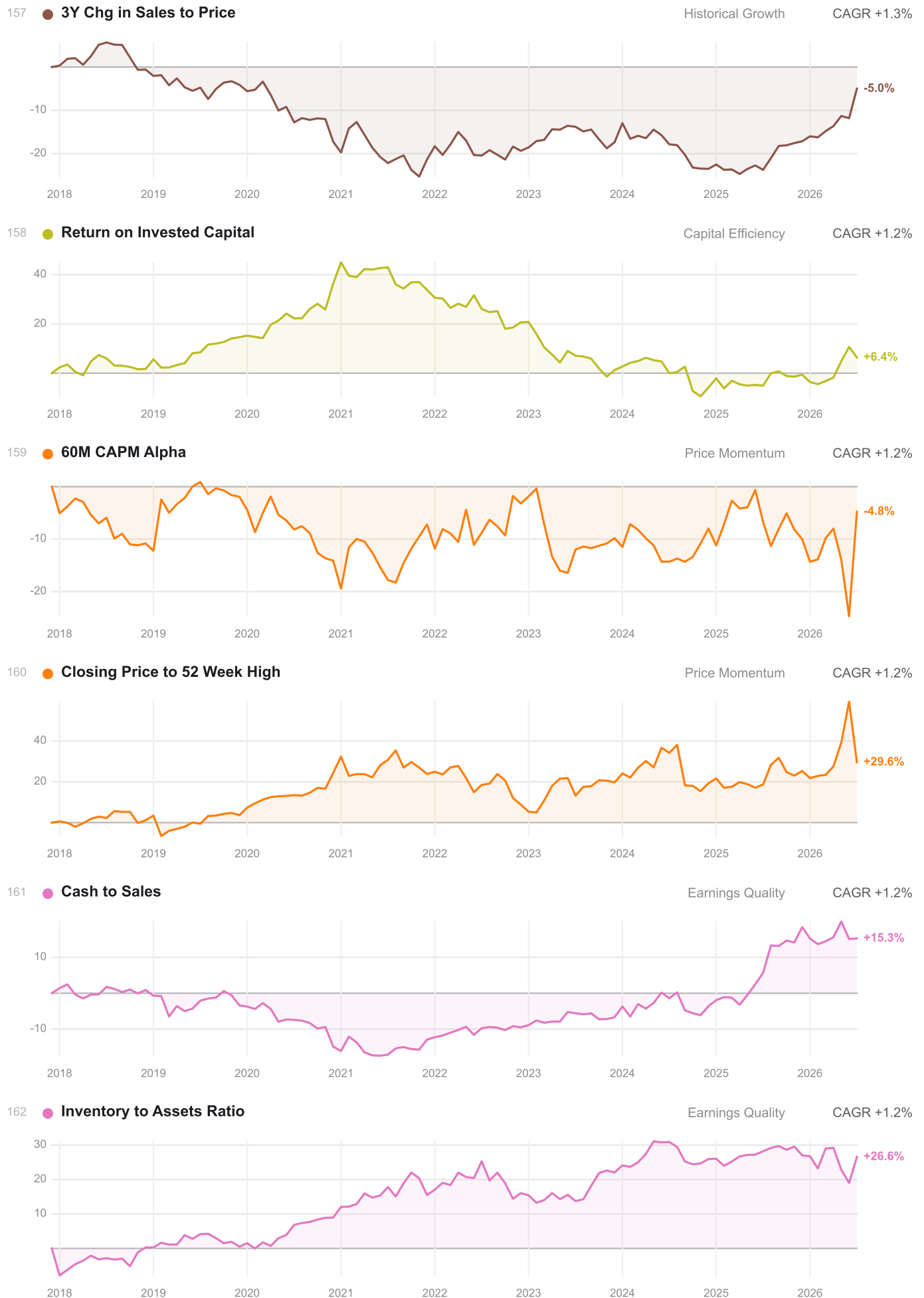
Size

CAGR +1.5%





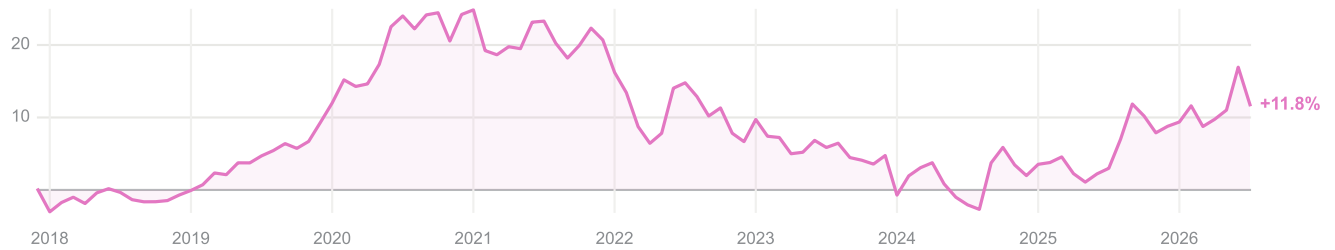




163 ● Quick Ratio

Earnings Quality

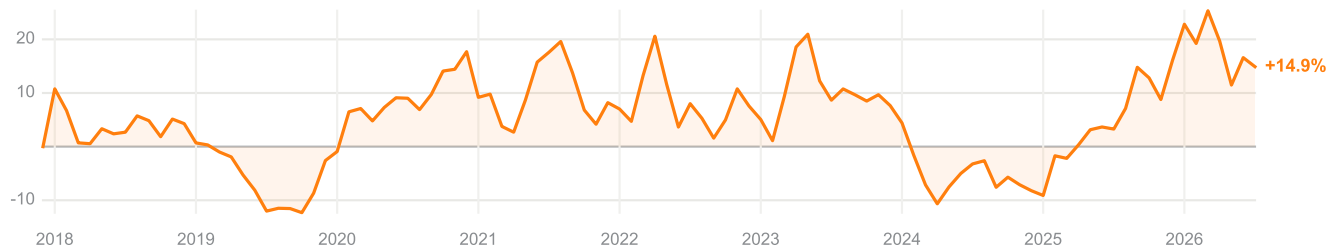
CAGR +1.1%



164 ● 1Y Chg in Share Turnover

Price Momentum

CAGR +1.1%



165 ● Log of Total Last Quarter Assets

Size

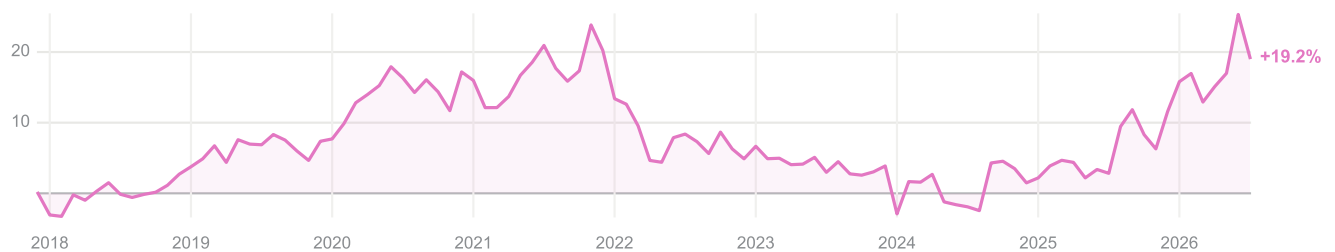
CAGR +1.1%



166 ● Current Ratio

Earnings Quality

CAGR +1.1%



167 ● Dividends to Cash Flow

Valuation

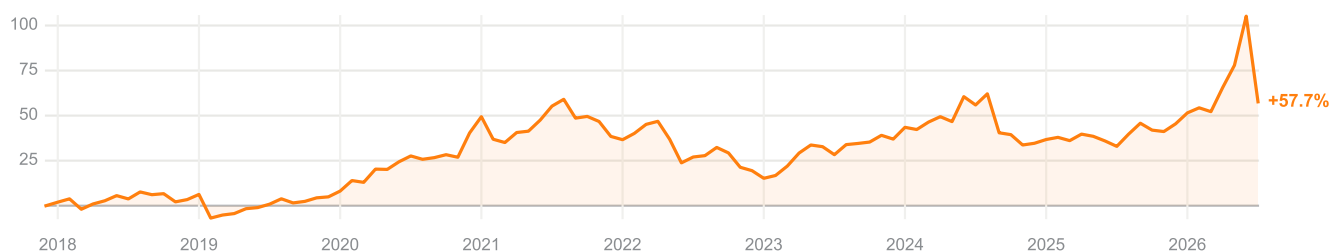
CAGR +1.1%

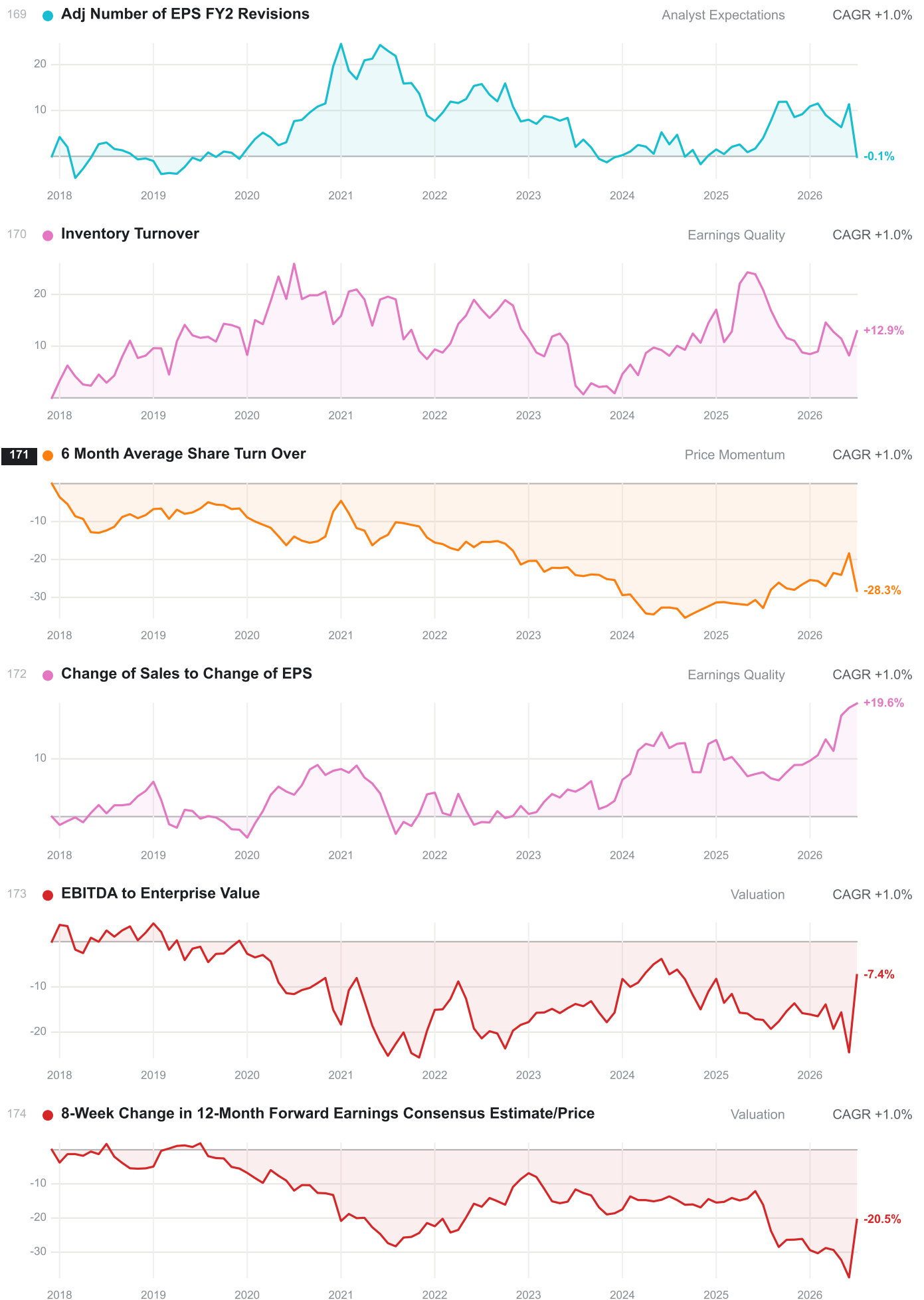


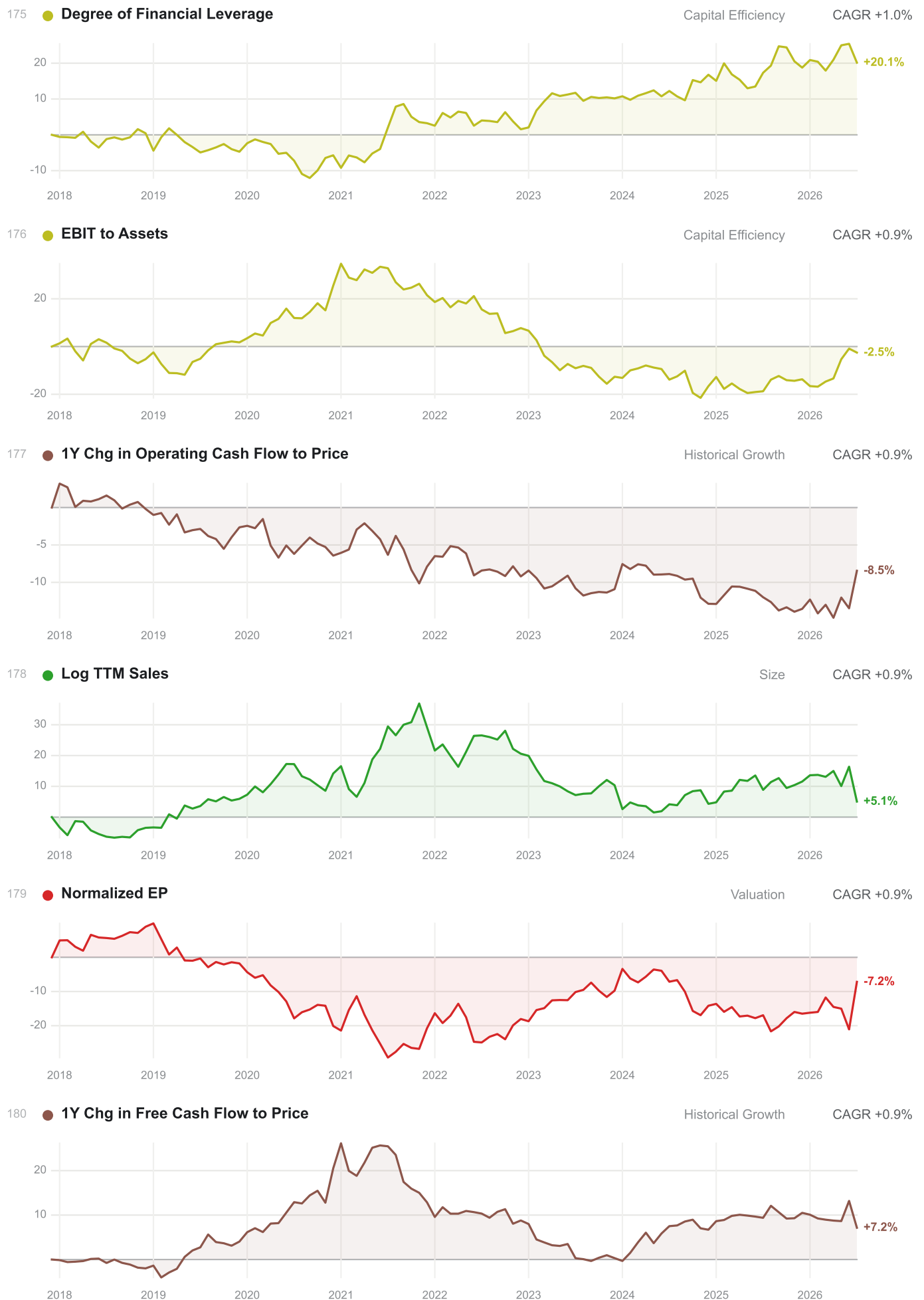
168 ● 4W to 52W Price Oscillator

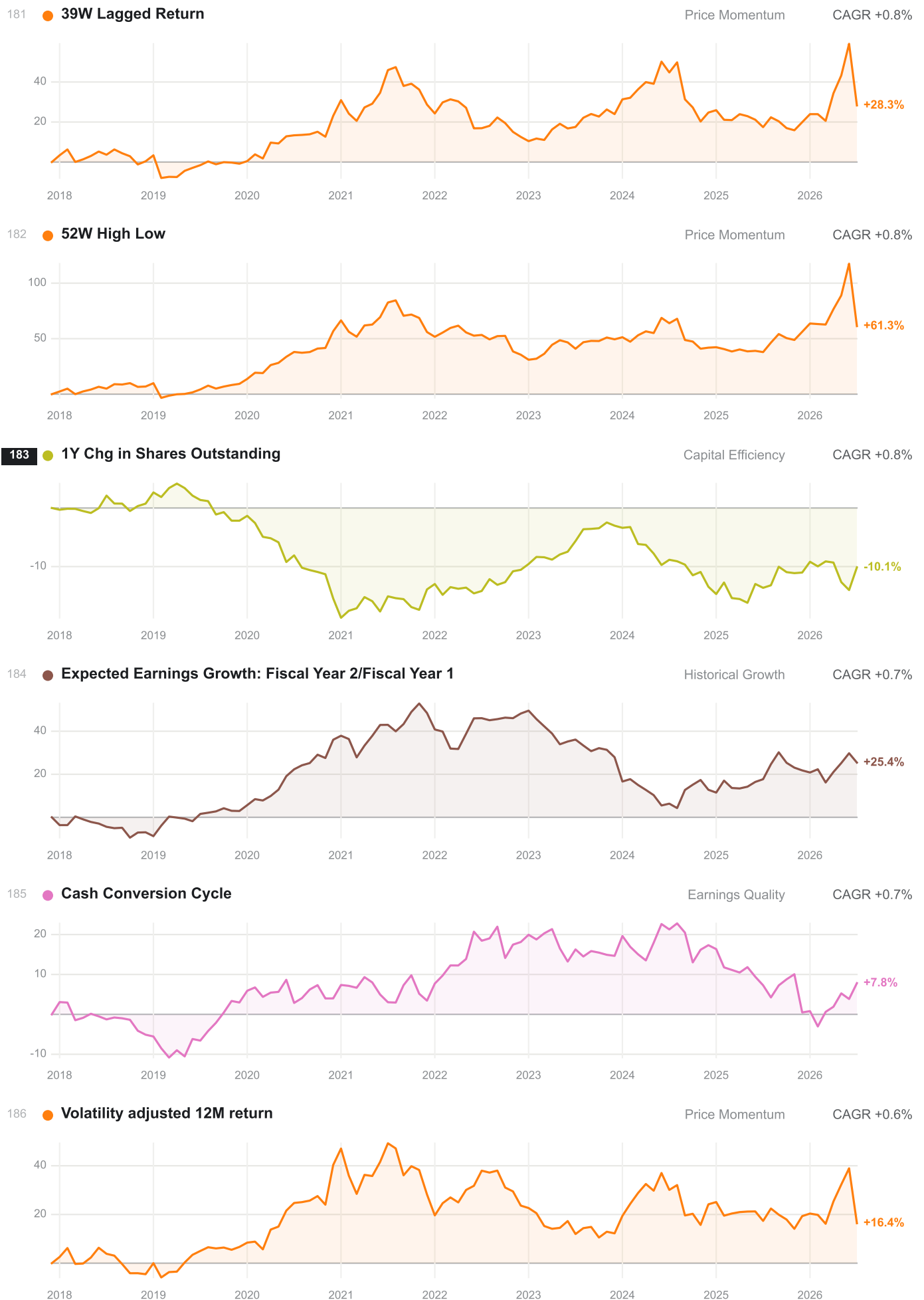
Price Momentum

CAGR +1.0%





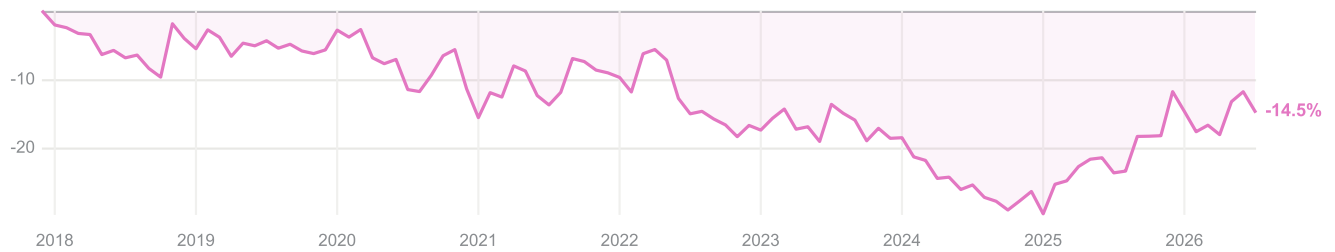




187 1Y Chg in Sales to Earnings

Earnings Quality

CAGR +0.6%

**188 Sustainable Growth Rate**

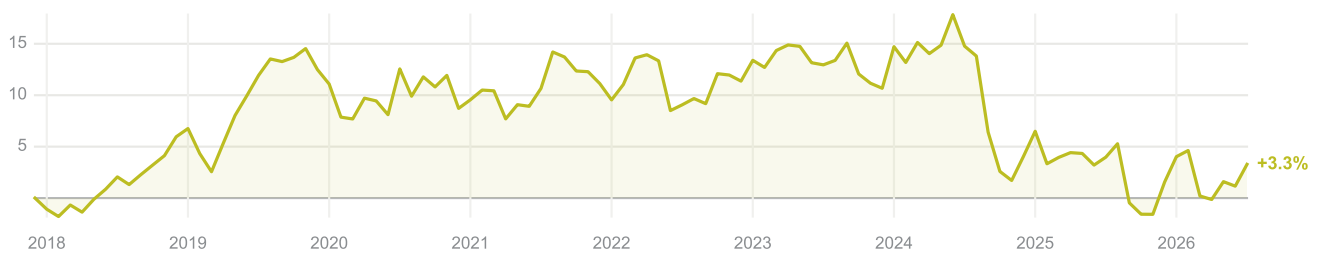
Historical Growth

CAGR +0.6%

**189 Year over Year Change of Total Debt**

Capital Efficiency

CAGR +0.5%

**190 Normalized ROE**

Capital Efficiency

CAGR +0.5%

**191 Capital Expenditure to Total Assets**

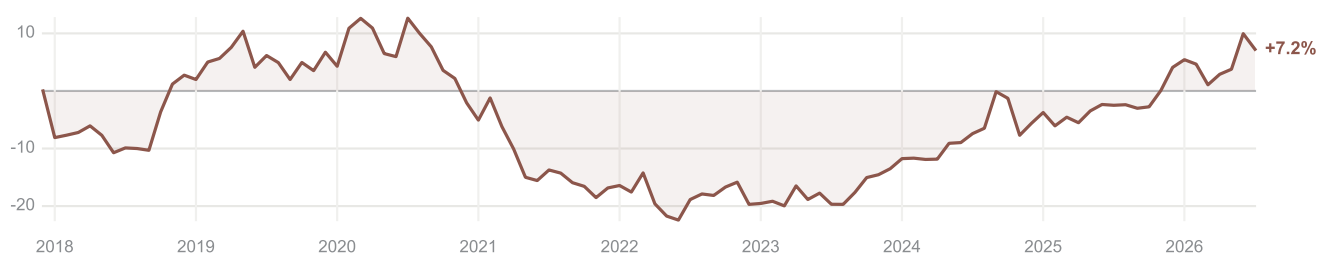
Capital Efficiency

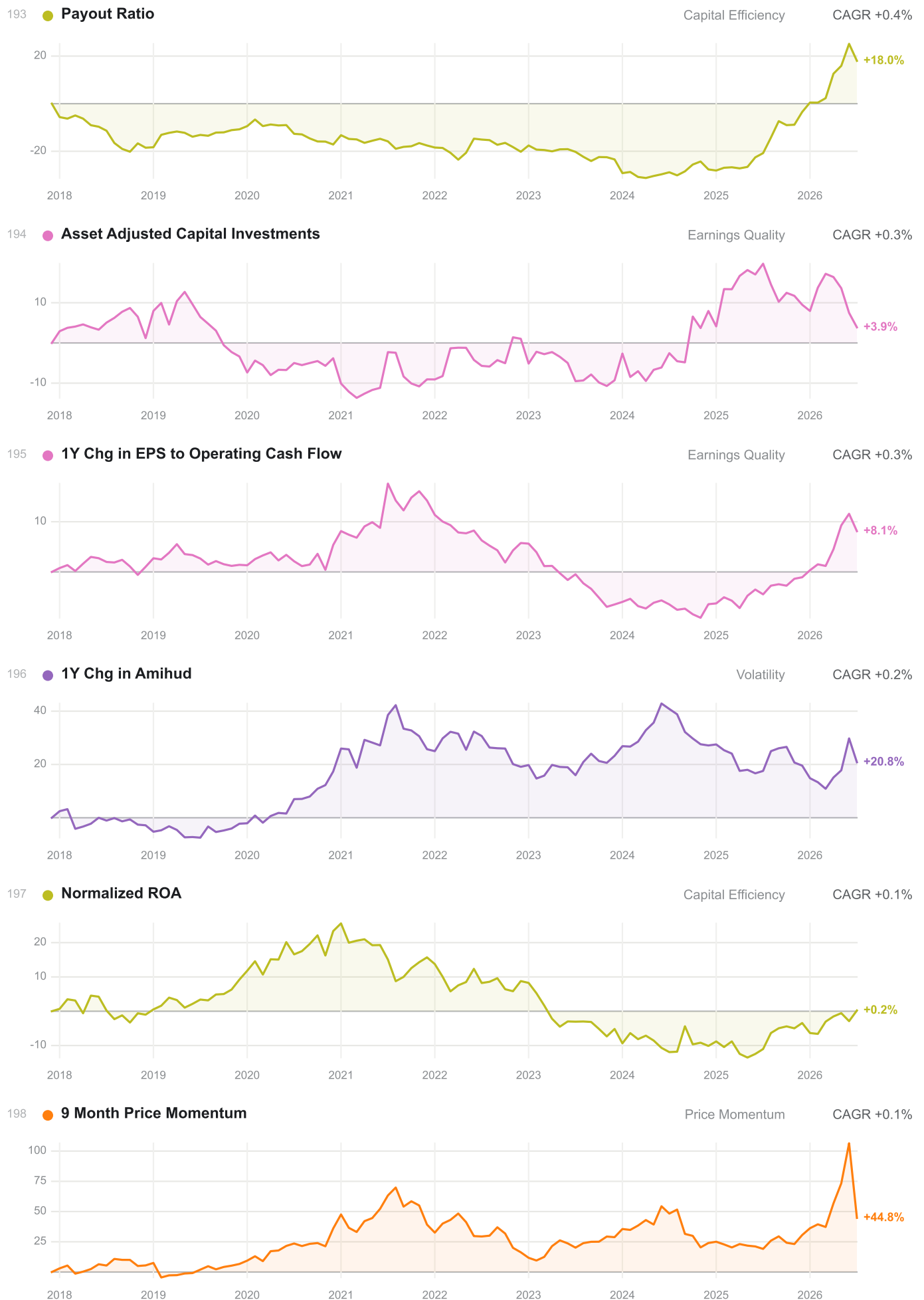
CAGR +0.5%

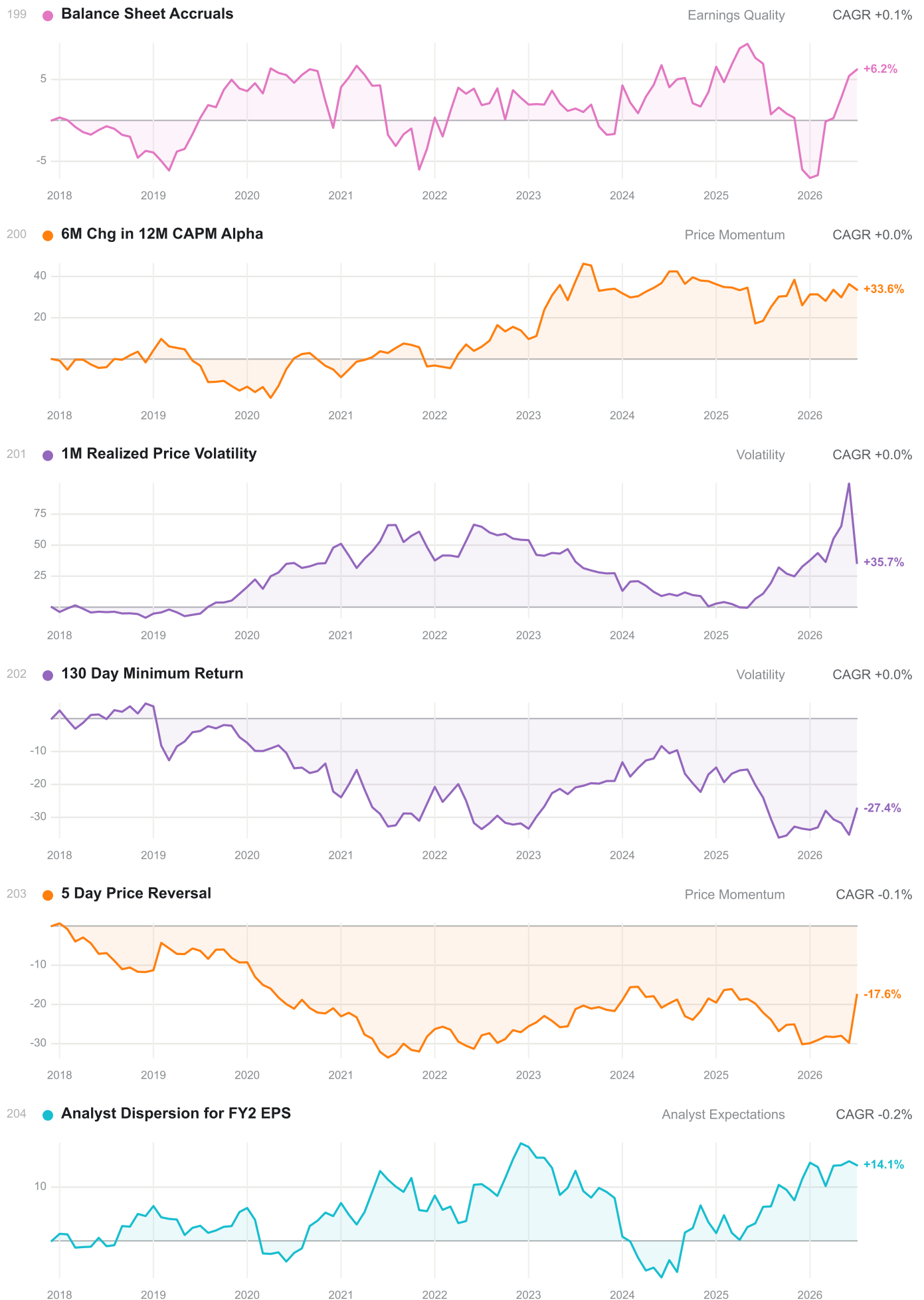
**192 Trailing 12-Month Overhead/Trailing 12-Month Sales**

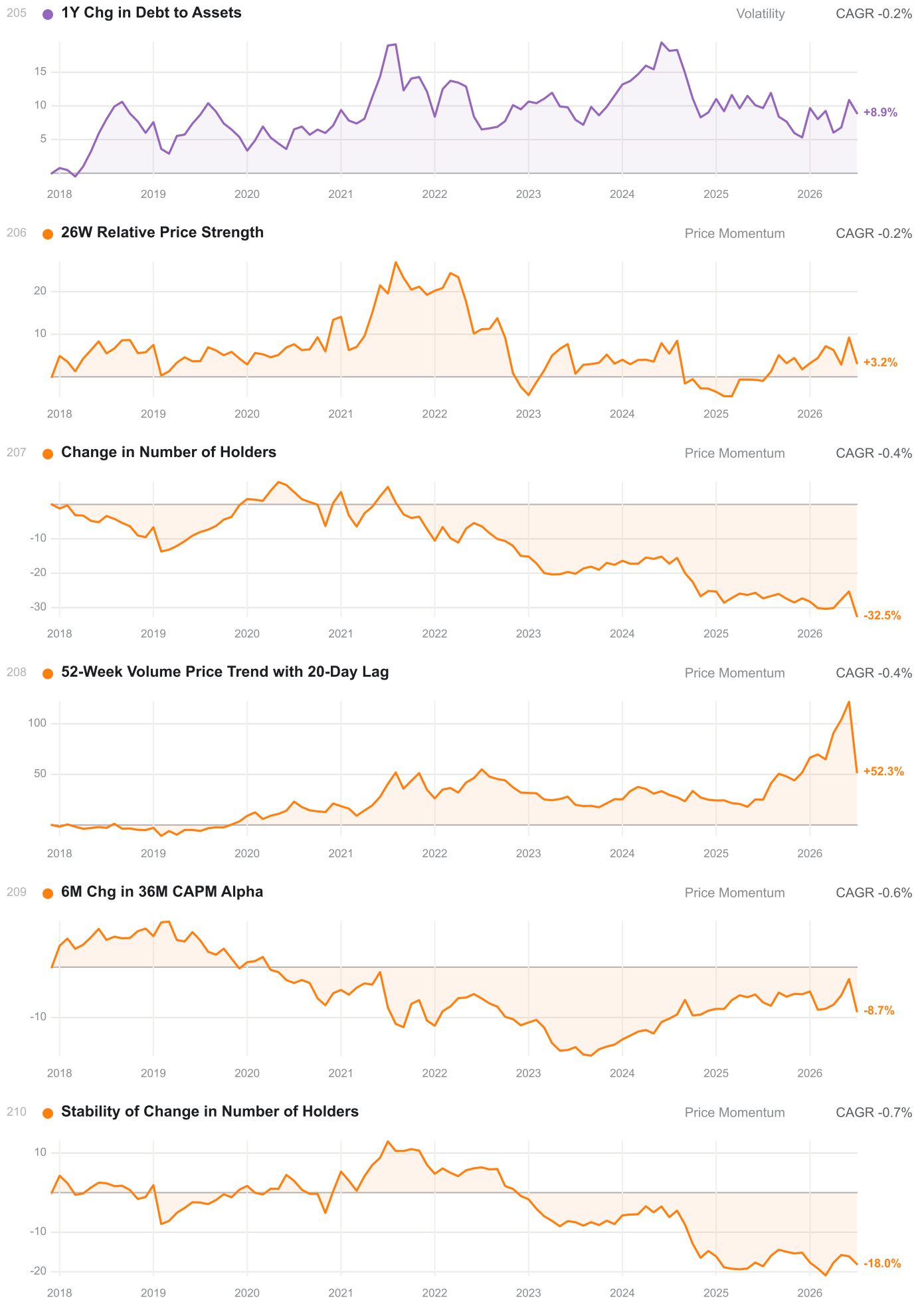
Historical Growth

CAGR +0.4%





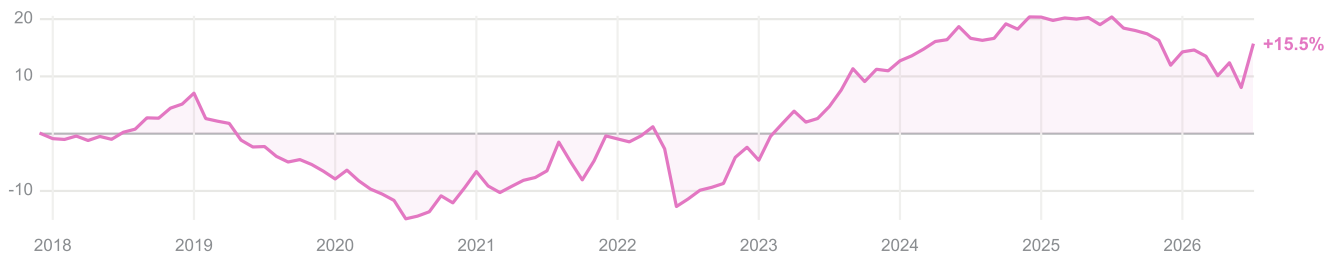




211 ● Sales to Gross Profit Margin Growth

Earnings Quality

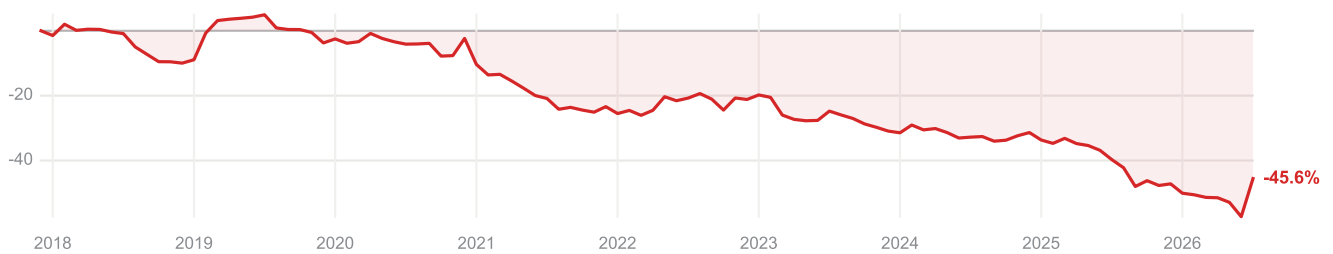
CAGR -0.7%



212 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

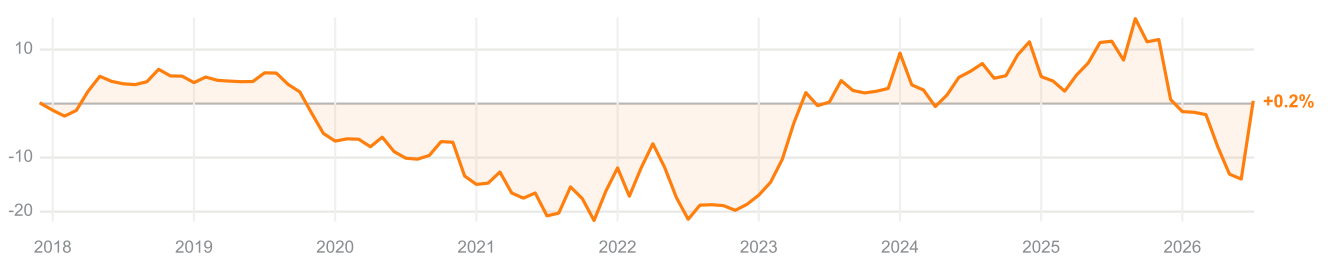
CAGR -0.7%



213 ● Institution Ownership Turnover

Price Momentum

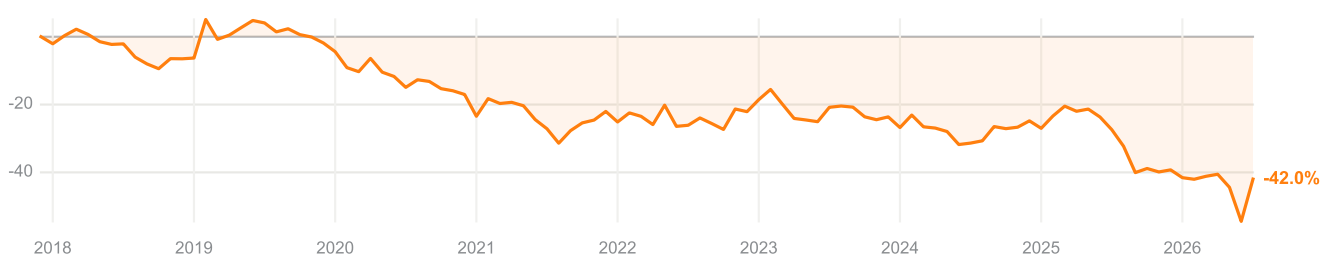
CAGR -0.8%



214 ● 1M Price Reversal

Price Momentum

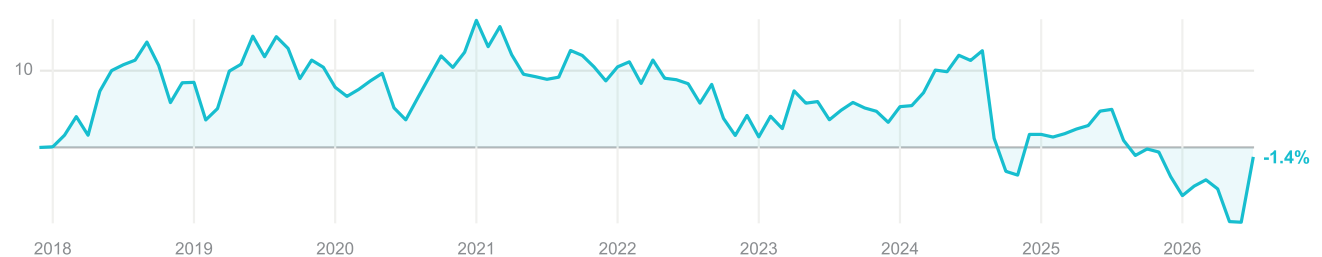
CAGR -0.8%



215 ● Revenue Estimate Dispersion

Analyst Expectations

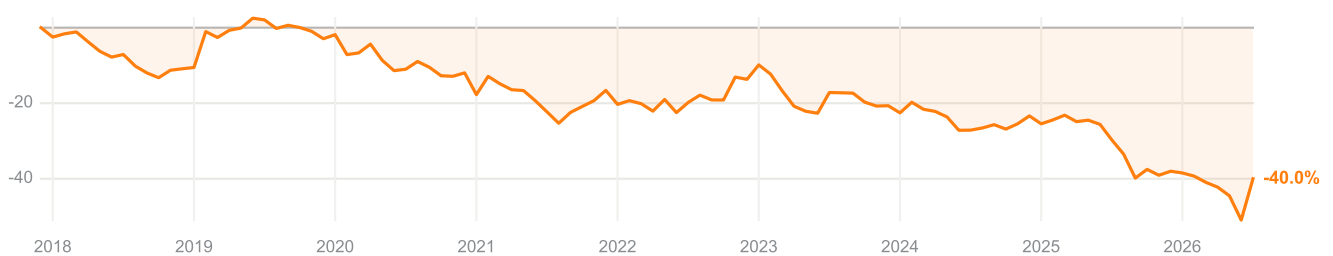
CAGR -1.0%



216 ● 14-Day Relative Strength Index

Price Momentum

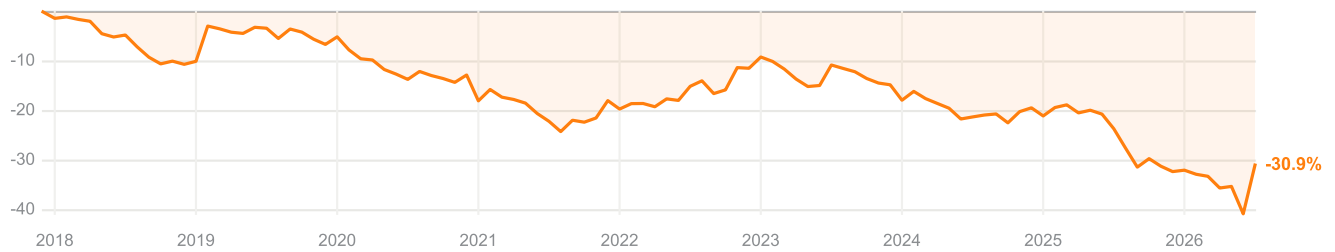
CAGR -1.1%



217 ● 1M Price High - 1M Price Low

Price Momentum

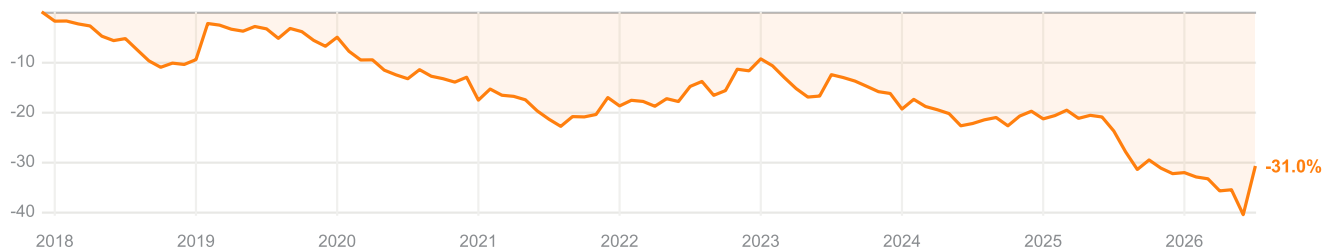
CAGR -1.1%



218 ● 20-Day Lane's Stochastic Indicator

Price Momentum

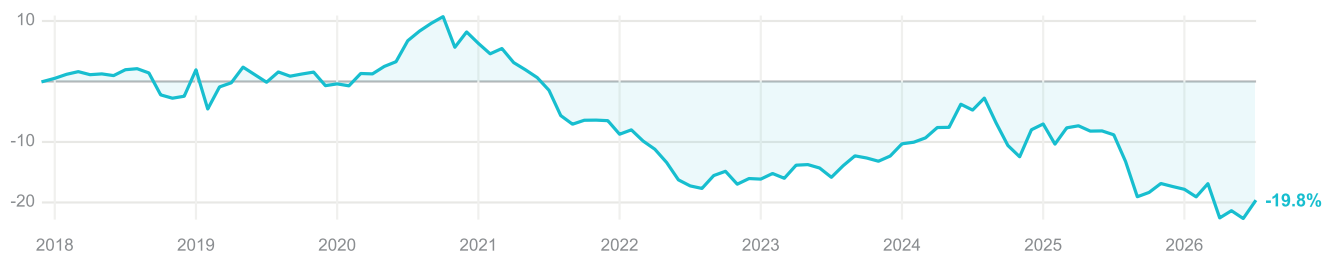
CAGR -1.2%



219 ● EBIT Estimate Dispersion

Analyst Expectations

CAGR -1.7%



220 ● 6M Chg in 18M CAPM Alpha

Price Momentum

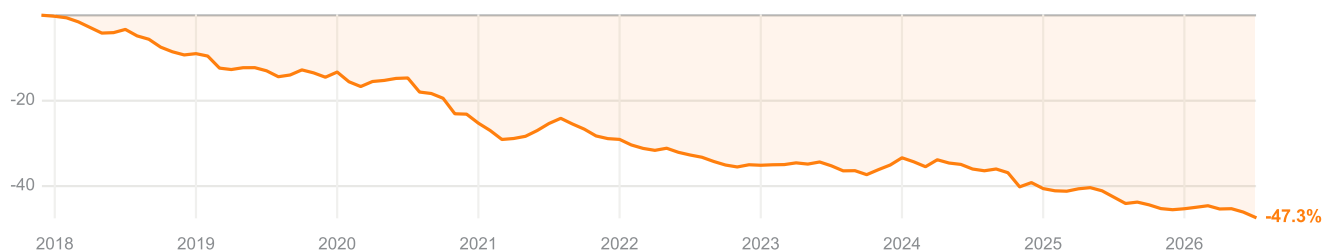
CAGR -2.0%



221 ● Change of Ownership Level - All Managers

Price Momentum

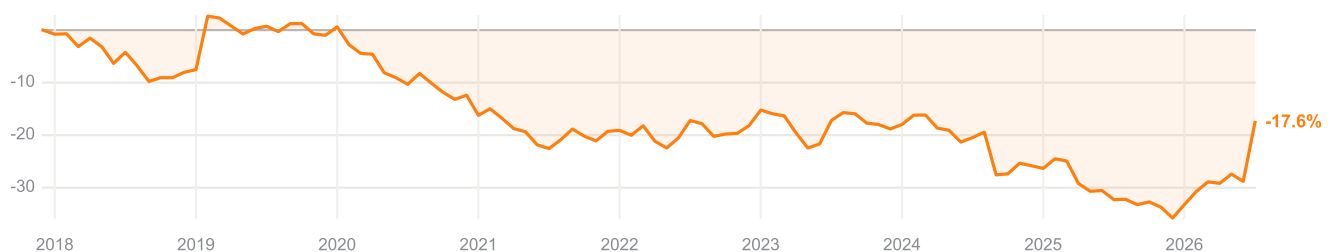
CAGR -2.2%



222 ● 5-Day Money Flow/Volume

Price Momentum

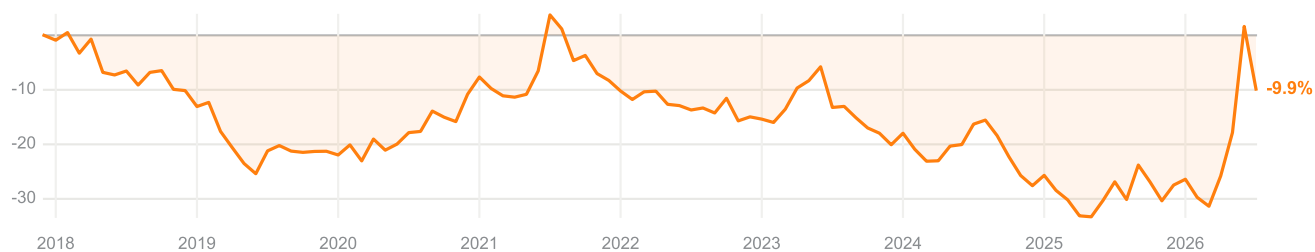
CAGR -2.6%



223 ● 5 Day Volume Signal

Price Momentum

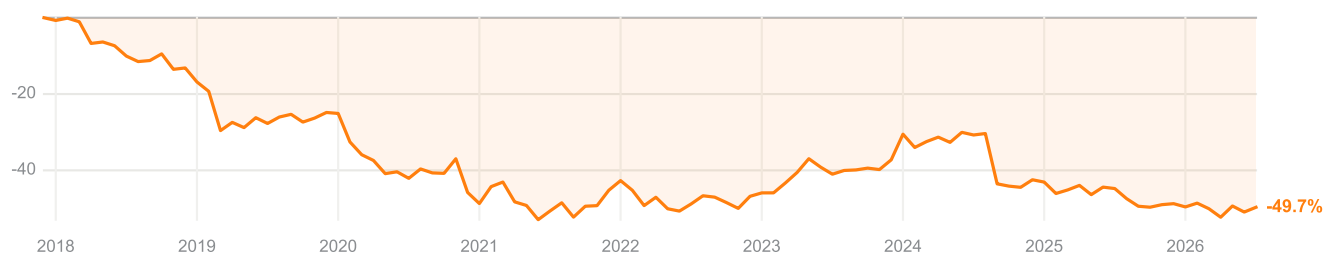
CAGR -4.5%



224 ● 20 Day Coefficient of Variation of Volume to Price

Price Momentum

CAGR -5.6%



225 ● 30 Day Coefficient of Variation of Volume to Price

Price Momentum

CAGR -6.0%

